

## Software Architecture Document (SAD)

### RERA Scrapper — MahaRERA Due-Diligence & Company Charter Pipeline

| Field                      | Value                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------------|
| <b>Document</b>            | Software Architecture Document                                                                       |
| <b>System</b>              | RERA Scrapper (MahaRERA project due-diligence and Company Charter generator)                         |
| <b>Repository root</b>     | RERA_Scrapper/                                                                                       |
| <b>Version</b>             | 1.1                                                                                                  |
| <b>Date</b>                | 13 August 2026                                                                                       |
| <b>Status</b>              | Baseline — describes the system as implemented                                                       |
| <b>Owner</b>               | Integrow Asset Management                                                                            |
| <b>Companion documents</b> | PRD.md (product requirements),<br>CLAUDE.md (flow), rules.md (content rules), guardrails.md (guards) |

### Table of contents

1. Purpose, scope and audience
2. Reference documents — the governance .md set
3. Architectural goals, principles and constraints
4. System context
5. Logical architecture — layered view
6. Module inventory
7. Runtime view — the end-to-end pipeline
8. Stage-by-stage specification
9. The Company Charter subsystem
10. Data architecture
11. The facts data model
12. Governance architecture — how rules.md is enforced
13. Guardrail architecture
14. Authentication and session architecture
15. External integration architecture
16. LLM integration architecture and cost model
17. Scoring subsystem
18. Document rendering architecture
19. Secondary and standalone builders
20. Concurrency and performance

21. [Failure, degradation and recovery model](#)
  22. [Security, privacy and compliance](#)
  23. [Testing architecture](#)
  24. [Deployment and operations](#)
  25. [Known limitations, technical debt and risks](#)
  26. [Appendices](#)
- 

## 1. Purpose, scope and audience

### 1.1 Purpose

This document describes the architecture of **RERA Scraper**: a single-command pipeline that takes a MahaRERA project registration number and produces a defensible, source-cited due-diligence pack — a project summary PDF plus a paired **Company Charter** (Internal and External variants) covering the counterparty, its promoters and the collateral.

It is the architectural reference for anyone extending, operating or auditing the system. It records not only *what* the components are, but *why* several of them exist in the shape they do, because a large fraction of this codebase is hard-won accommodation of a live government portal that does not behave the way its API surface suggests.

### 1.2 Scope

In scope:

- The full ingestion – enrichment – assembly – rendering – verification pipeline.
- Both entry points: `main.py` (CLI) and `app.py` (Streamlit).
- The three CAPTCHA-gated external portals and the redundancy chains built around them.
- The rules/guardrail governance layer that decides whether a document may ship.
- The two document families (RERA summary PDF; Company Charter family) and three renderers.

Out of scope:

- The commercial underwriting decision the documents feed into.
- Any MahaRERA-side system internals (treated as an opaque, occasionally hostile third party).
- CI/CD infrastructure — none currently exists (see §25).

### 1.3 Audience

| Audience                           | Read                                      |
|------------------------------------|-------------------------------------------|
| New engineer joining the codebase  | §3–§9, then <code>CLAUDE.md</code>        |
| Engineer changing a guard or gate  | §12, §13, then <code>guardrails.md</code> |
| Engineer changing document content | §11, §12, §18, then <code>rules.md</code> |

| Audience                    | Read                        |
|-----------------------------|-----------------------------|
| Operations / runbook author | \$14, \$21, \$24            |
| Reviewer / auditor          | \$12, \$13, \$16, \$22      |
| Product                     | PRD.md first, then \$4, \$7 |

## 2. Reference documents — the governance .md set

This repository is unusual in that **some of its Markdown is executable configuration, not prose**. Understanding which is which is a prerequisite to changing anything.

|                                              |  |                     |                 |               |           |
|----------------------------------------------|--|---------------------|-----------------|---------------|-----------|
| +-----+<br>  THE MARKDOWN GOVERNANCE SET<br> |  |                     |                 |               |           |
| +-----+                                      |  |                     |                 |               |           |
| README.md                                    |  | CLAUDE.md           |                 | rules.md      |           |
| +-----+                                      |  | +-----+             |                 |               |           |
| +-----+                                      |  |                     |                 |               |           |
| Human onboarding                             |  | THE FLOW            |                 | THE           |           |
| CONTENT                                      |  |                     |                 |               |           |
| Usage, install,                              |  | What runs, in       |                 | RULES         |           |
|                                              |  |                     |                 |               |           |
| ASCII pipeline                               |  | ----->  what order, | ----->          | 3             |           |
| sections,                                    |  |                     |                 |               |           |
| architecture                                 |  | "how"               | what is opt-in, | "when"        | A / B / C |
|                                              |  |                     |                 |               |           |
| Output layout                                |  | what may not be     |                 |               |           |
|                                              |  |                     |                 |               |           |
|                                              |  | skipped             |                 | *** PARSED    |           |
| AT                                           |  |                     |                 |               |           |
| Status: prose                                |  |                     |                 |               |           |
| RUNTIME ***                                  |  |                     |                 |               |           |
| +-----+                                      |  | Status: prose       |                 | Status:       |           |
| LIVE                                         |  |                     |                 |               |           |
|                                              |  | +-----+             |                 |               |           |
| CONFIG                                       |  |                     |                 |               |           |
|                                              |  |                     |                 | +-----        |           |
| +-----+                                      |  |                     |                 |               |           |
|                                              |  | v                   |                 |               | v         |
|                                              |  | guardrails.md       |                 |               |           |
| _read_rules_section()                        |  | +-----+             |                 | in            |           |
| company_charter.py                           |  | THE GUARDS          |                 | injects B and |           |
| C into                                       |  |                     |                 |               |           |

|             |                 |              |
|-------------|-----------------|--------------|
| calls       | Every gate,     | Claude API   |
|             | fallback, bound |              |
| validated   | and never-fatal | Section A is |
|             | wrapper, BY     | but NEVER    |
| transmitted | SYMBOL          |              |
|             | Status: prose,  |              |
|             | but test_guard- |              |
|             | rails_doc.py    |              |
|             | fails if a      |              |
|             | named symbol    |              |
|             | disappears      |              |
|             | +-----+         |              |

CHARTER\_RESHAPE\_SPEC.md  
 .claude/skills/maharera-report/SKILL.md

|                       |                                                           |
|-----------------------|-----------------------------------------------------------|
| +-----+               | +-----+                                                   |
| HISTORICAL RECORD     | Project-local Claude Code skill                           |
| COMPLETED 2026-08-10  | for driving report generation                             |
| *** DO NOT FOLLOW *** | interactively                                             |
| Stale line refs,      | +-----+                                                   |
| stale dead-code claim |                                                           |
| +-----+               | output/company_charters/<br>CHARTER_RESHAPE_CHANGE_LOG.md |
|                       | +-----+                                                   |
|                       | Per-change log of the 2026-08                             |
|                       | Charter reshape                                           |
|                       | +-----+                                                   |

| File          | Size     | Role                                                                                                              | Mutability                                                                                                                 |
|---------------|----------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| CLAUDE.md     | ~5.5 KB  | Pipeline flow map; the binding <b>flow</b> authority                                                              | Prose; edit when the flow changes                                                                                          |
| rules.md      | ~18.4 KB | Content rules; the binding <b>content</b> authority. Sections B and C are injected verbatim into Claude API calls | <b>Live configuration.</b> Parsed by <code>_read_rules_section()</code> . Section markers must not be renamed or reordered |
| guardrails.md | ~5.6 KB  | Map of every gate, fallback, bound and never-fatal wrapper, keyed <b>by symbol</b> , <b>never by line</b>         | Prose, but <code>test_guardrails_doc.py</code> fails the suite if a named symbol stops existing                            |

| File                                                                  | Size     | Role                                                                                                        | Mutability  |
|-----------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------|-------------|
| README.md                                                             | ~15.8 KB | <b>number</b><br>Onboarding, usage,<br>the original ASCII<br>pipeline architecture,<br>output layout        | Prose       |
| CHARTER_RESHAPE_<br>SPEC.md                                           | ~13.1 KB | <b>Historical only.</b> All<br>seven tasks shipped<br>2026-08-10. Explicitly<br>stale in four named<br>ways | Frozen      |
| .claude/skills/<br>maharera-<br>report/SKILL.md                       | ~8.3 KB  | Project-local agent<br>skill for the report<br>workflow                                                     | Living      |
| output/<br>company_charters<br>/<br>CHARTER_RESHAPE_<br>CHANGE_LOG.md | ~8.9 KB  | Change log of the<br>reshape                                                                                | Append-only |

**Precedence rule, quoted from rules.md:** “Where the two disagree, this file wins on content and CLAUDE.md wins on flow.”

## 2.1 The three sections of rules.md

```
rules.md
+-----+
| --- Section A: CODING-TIME RULES --- |
| Audience: engineers.                  |
| Delivery: read by humans; validated by _preflight_rules. |
| *** NEVER SENT TO AN API. ***        |
| test_claude_md_doc_review.py asserts its absence. |
+-----+
| --- Section B: COMMON CONTENT RULES --- |
| Audience: the model.                  |
| Delivery: _common_content_rules() -> a separate cacheable |
| system-prompt block on EVERY content call, BOTH variants. |
| Constraint: must contain no em dash and no " -- ". |
+-----+
| --- Section C: STAGE-FIXED CONTENT RULE (external only) --- |
| Audience: the model, external-facing calls only. |
| Delivery: _external_citation_rule(), appended after B. |
| Same punctuation constraint as B. |
+-----+
```

The punctuation constraint on B and C is not stylistic.

\_verify\_external\_document\_quality hard-fails an External save containing an em dash or a hyphen-pair dash; prompt punctuation bleeds into model output; therefore the prompt

itself must be clean or every External save fails. `_preflight_rules` enforces this before a paragraph is written.

---

### 3. Architectural goals, principles and constraints

#### 3.1 Goals

| #  | Goal                                                    | How the architecture serves it                                                                                                                    |
|----|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| G1 | <b>Every claim is traceable to a source</b>             | <code>_FIELD_WITH_SOURCE</code> schema shape; <code>sources[]</code> with topic enumeration; per-claim citation resolution; source-trust registry |
| G2 | <b>A partial failure never costs the run</b>            | Never-fatal wrappers around every optional stage; per-category thread isolation; manifest-row failure recording                                   |
| G3 | <b>A bad document must not ship</b>                     | Three-tier gate chain: preflight → mechanical gate → semantic review, the last of which blocks the PDF                                            |
| G4 | <b>A model can only match or improve, never degrade</b> | Every model-backed judgement keeps its deterministic predecessor behind it; cache-miss is indistinguishable from “no model ran”                   |
| G5 | <b>The record keeps what the page drops</b>             | Scrub/restore contract before <code>.facts.json</code> is written                                                                                 |
| G6 | <b>Re-runs are cheap and diffable</b>                   | Run archiving, document reuse by <code>(document_id, filename)</code> , 24-hour research reuse window                                             |
| G7 | <b>Nothing is silently guessed</b>                      | Absent data becomes a gap, never an approximation; disagreeing sources are both surfaced, never silently reconciled                               |

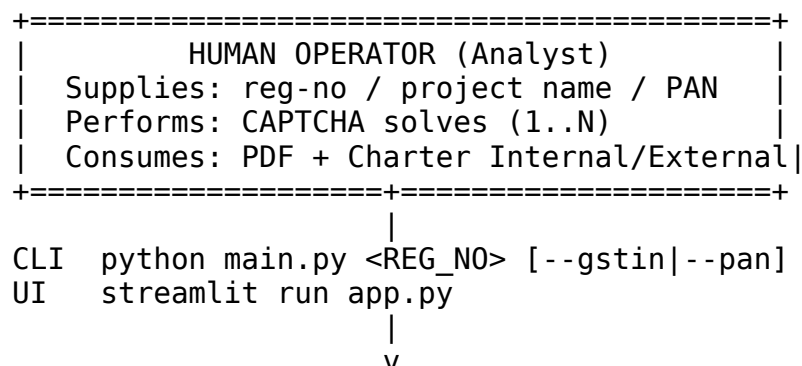
### 3.2 Principles

1. **Deterministic core, probabilistic edges.** Rendering is pure code. The model assembles facts and offers judgements; it never renders the External document.
2. **Fail-soft on ingestion, fail-hard on output.** Getting less data is acceptable; shipping a wrong document is not.
3. **Human-in-the-loop where the law puts a CAPTCHA.** No module reads or solves a CAPTCHA image. This is a hard constraint, stated in `session_auth.py`, `gst_portal.py` and `mahabhumi.py`.
4. **Symbols, not line numbers.** `guardrails.md` records guards by `module.symbol`; line references in this repo have gone stale twice.
5. **Two contradictory numbers beat one invented one.** Two known data contradictions are deliberately left unreconciled (`rules.md` §A).
6. **Absence is not a finding.** The single largest editorial rule: a clean check produces no sentence at all.

### 3.3 Constraints

| Constraint                                                                           | Consequence                                                                            |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| MahaRERA gates 7 of 9 categories behind a browser CAPTCHA                            | A human must be present for a full run, or supply a token                              |
| Maha Bhulekh regenerates its CAPTCHA on every partial postback                       | No reusable session; land-record lookup is opt-in and manual                           |
| GST portal needs one solve per PAN search <b>and</b> one per GSTIN                   | GST intake is opt-in and placed beside the other browser work                          |
| docx2pdf is Windows-only (Word COM automation)                                       | PDF conversion — and therefore the deliverable — requires Windows                      |
| output/ is gitignored, and holds the .docx template                                  | Template must be backed up manually before structural change                           |
| No packaging manifest ( <code>pyproject.toml</code> / <code>setup.py</code> ) exists | Flat module namespace at repo root; <code>requirements.txt</code> is the only manifest |
| Sonnet 5 intro pricing expires <b>2026-08-31</b>                                     | Cost figures under-report after that date unless updated                               |

## 4. System context



```
+=====+
|                                     RERA  SCRAPPER  (this system)
|
|                                     Python 3.11+, single process, local disk
|
+=====+
+-----+ +-----+ +-----+ +-----+ +-----+
|         |         |         |         |         |
|         v         v         v         v         v
+-----+ +-----+ +-----+ +-----+ +-----+
| MahaRERA   | MCA      | Credit /   | GST       | Anthropic  |
| (2 hosts)  | mirrors | IBBI      | Portal    | Claude API |
| maharera.  | Zauba-  | ICRA      | services. | claude-    |
| maha...    | Corp    | Infomer-  | gst.gov   | sonnet-5   |
| (Drupal,   | Tofler  | ics       | .in       | + web_search|
| search)    | Insta-  | IBBI      |           | server     |
| maharerait | Financials |          | CAPTCHA  | tool       |
| .maha...   | cials    | (public   | per       | ANTHROPIC_ |
| (API,      | (3-way   | search)   | lookup    | API_KEY    |
| CAPTCHA)   | cross-   |           |           |            |
+-----+ +-----+ +-----+ +-----+ +-----+
|         |         |         |         |         |
|         v         v         v         v         v
+-----+ +-----+ +-----+ +-----+ +-----+
+
| Maha      | | Open      | | Google    | | xAI        | | Tesseract  |
| Bhulekh   | | Street    | | Maps      | | Grok       | | OCR (local)|
| (land     | | Map       | | (opt-in,  | | (opt-in   | | eng + mar  |
| records,  | | Nominati | | ToS       | | fallback  | | traineddata|
| Marathi,  | | (geocode | | caveat)   | | verifier  | | bundled    |
| CAPTCHA   | | 1 req/s) | |           | | XAI_API_| |            |
| per call) | |          | | COMPANY_  | | KEY)      | | TESSERACT_ |
+-----+ +-----+ | CHARTER_  | +-----+ | CMD
|                               | USE_MAPS_ | +-----+
+
```



```
| SCRAPE=1 |  
+-----+
```

LEGEND: == system boundary      -- component boundary  
         --> synchronous call      v data flow direction

**Trust tiering of the external world.** The system does not treat all sources equally. `_classify_source_tier` maps every source into one of 12 tiers with weights 20–100, feeding the Documentation Confidence score. MahaRERA, MCA mirrors, rating agencies, IBBI and Maps are *trusted by design* and excluded from the open-web trust registry; everything else (99acres, NoBroker, press, social, Wikipedia) accumulates hit counts in `source_trust_registry.json` across runs and is auto-promoted at the **5th distinct project**.

---

## 5. Logical architecture — layered view

```
+=====+  
====+  
| L6 PRESENTATION / DELIVERY  
|  
| report.py (ReportLab PDF) . company_charter._fill_template (docx)  
|  
| charter_report.py (docx) . executive_briefing.py (docx +  
matplotlib) |  
| _convert_docx_to_pdf (docx2pdf / Word COM) -> THE DELIVERABLE IS  
PDF |  
+=====+  
====+  
|  
| L5 GOVERNANCE / VERIFICATION  
|  
| _preflight_rules . _verify_external_document_quality (11 checks)  
|  
| run_claude_md_document_review + _verified_violations [BLOCKS THE  
PDF] |  
| charter_report.verify_charter_report_quality (6 checks)  
|  
| rules.md sections A / B / C . guardrails.md symbol map  
|  
+=====+  
====+  
|  
| L4 EDITORIAL / NORMALISATION  
|  
| run_editorial_passes (3 model passes, all with deterministic
```

```

fallback) |
| _scrub_clean_checks . _sanitize_process_gaps .
| normalize_misfiled_facts|
| _classify_flags . _flag_headline . run_finding_research
|
| _restore_clean_checks -> the record keeps what the page drops
|
+=====
====+
^

+=====
====+
| L3 ANALYSIS / SCORING
|
| _compute_developer_score (3 buckets, 9 sub-metrics, no
renormalisation)|
| _compute_documentation_confidence_score (8 criteria, renormalises)
|
| gst_compliance (statutory due dates, friction-point model)
|
| verify_cross_corroboration . _record_source_hits_and_promote
|
+=====
====+
^

+=====
====+
| L2 ENRICHMENT
|
| deep_research.py (agentic web research) . promoter_portfolio.py
|
| MCA-mirror chain (_run_mca_profile_chain) . credit rating .
IBBI |
| gst_intake / gst_portal . mahabhumi (CTS land records)
|
| search_maharera_judgments . run_review_authenticity_triage
|
+=====
====+
^

+=====
====+
| L1 INGESTION
|
| resolver.py (Playwright search) . session_auth.py (CAPTCHA ->
token) |
| token_cache.py (90-min disk cache) . api_client.py (9 categories,
|
| document + complaint-order download) . discover.py (--verify)
|

```

```

| run_archive.py (snapshot previous run)
|
+=====
====+
                                ^
+=====
====+
| L0 FOUNDATION
|
| config.py (endpoints, selectors, timeouts, caps)
|
| Local filesystem: output/ tree . source_trust_registry.json
|
| .guest_token_cache.json . tessdata/
|
+=====
====+

```

**Cross-cutting:** cost/usage accounting (`deep_research._record_usage`, per-label) spans L2–L5; the never-fatal wrapper policy spans L1–L5; the facts dict is the shared data structure from L2 upward.

---

## 6. Module inventory

### 6.1 Application modules (29 files, all at repository root)

| Module                       | LOC/size | Layer         | Role                                                                             |
|------------------------------|----------|---------------|----------------------------------------------------------------------------------|
| <code>main.py</code>         | 25.3 KB  | Orchestration | CLI entry point; owns the 11-stage sequence and the never-fatal wrappers         |
| <code>app.py</code>          | 28.0 KB  | Presentation  | Streamlit UI over the same functions; 3 tabs; no duplicated logic                |
| <code>config.py</code>       | 5.8 KB   | L0            | Endpoints, selectors, timeouts, CATEGORY_ORDER, PROMOTER_PROJECT_LIMIT           |
| <code>resolver.py</code>     | 8.2 KB   | L1            | reg-no / name → internal numeric project_id via Playwright; also promoter search |
| <code>session_auth.py</code> | 3.8 KB   | L1            | Visible browser,                                                                 |

| Module                | LOC/size | Layer | Role                                                                                       |
|-----------------------|----------|-------|--------------------------------------------------------------------------------------------|
|                       |          |       | human CAPTCHA solve, reads accessToken from sessionStorage                                 |
| token_cache.py        | 2.7 KB   | L1    | 90-minute disk token cache + standalone CLI                                                |
| api_client.py         | 26.0 KB  | L1    | 9 category endpoints, envelope normalisation, document + complaint-order download          |
| discover.py           | 1.8 KB   | L1    | --verify mode: probe every endpoint, report confirmed-vs-observed drift                    |
| run_archive.py        | 5.8 KB   | L1    | Snapshot the previous run and the previous Charter; supply reusable pieces                 |
| promoter_portfolio.py | 24.3 KB  | L2    | Promoter's whole RERA portfolio, complaint/appeal totals, 5 km area, geocoding             |
| deep_research.py      | 38.5 KB  | L2    | <b>The single transport for every Claude API call;</b> agentic research; usage/cost ledger |
| gst_intake.py         | 9.9 KB   | L2    | PAN → all GSTINs → filing tables → gst_filing_input.json                                   |
| gst_portal.py         | 18.5 KB  | L2    | GST portal driver + filing-table parser; OCR-assisted                                      |
| gst_compliance.py     | 12.2 KB  | L3    | Pure offline GSTIN/PAN validation, QRMP                                                    |

| Module                    | LOC/size                       | Layer      | Role                                                                                                                 |
|---------------------------|--------------------------------|------------|----------------------------------------------------------------------------------------------------------------------|
| mahabhumi.py              | 32.9 KB                        | L2         | frequency, statutory due dates, delays<br>Maha Bhulekh Property Card lookup; Marathi UI; district map; OCR mar+eng   |
| cts_intake.py             | 3.0 KB                         | Entry      | Standalone CTS lookup with no RERA number                                                                            |
| cts_resolve.py            | 6.7 KB                         | Entry      | 4-step human-in-the-loop CTS office/village/number resolution                                                        |
| promoter_intake.py        | 3.0 KB                         | Entry      | Standalone CIN-only promoter checks                                                                                  |
| attach_rera.py            | 2.3 KB                         | Entry      | Link a pending CIN/CTS case to a RERA number                                                                         |
| <b>company_charter.py</b> | <b>591.9 KB / 10,182 lines</b> | L2–L6      | <b>The core.</b> Facts assembly, registry chains, editorial passes, scoring, dual-variant rendering, all three gates |
| charter_report.py         | 70.8 KB                        | L6         | Second builder — “Counterparty + Collateral” document; own quality gate                                              |
| charter_document.py       | 32.4 KB                        | Shared lib | Builder deleted 2026-08-10; <b>still a live shared library</b> (16 symbols)                                          |
| charter_research_prep.py  | 24.6 KB                        | Shared lib | Roster building + agent prompt construction for the charter_report pipeline                                          |
| run_charter_pipeline.py   | 5.4 KB                         | Entry      | prepare() / finish() seam for the agent-driven                                                                       |

| Module                | LOC/size | Layer   | Role                                                                         |
|-----------------------|----------|---------|------------------------------------------------------------------------------|
| executive_briefing.py | 21.0 KB  | L6      | charter report<br>8-page narrative companion; donut charts via matplotlib    |
| report.py             | 26.5 KB  | L6      | The RERA project summary PDF (ReportLab)                                     |
| build_report.py       | 11.8 KB  | One-off | Hand-curated research for one engagement; “delete or rewrite per engagement” |
| finalize_report.py    | 4.6 KB   | Entry   | Rebuild the summary PDF from disk with <b>zero network calls</b>             |

## 6.2 Test modules (28 files, test\_\*.py at root)

Grouped by what they protect:

| Group                                         | Files                                                                                                                                                                                                  |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Guard integrity</b> (stop a guard rotting) | test_guardrails_doc.py,<br>test_rules_preflight.py,<br>test_claude_md_doc_review.py,<br>test_claude_md_and_charter_fixes.py                                                                            |
| <b>Editorial policy</b>                       | test_editorial_passes.py,<br>test_clean_check_scrubber.py,<br>test_process_text_sanitizer.py,<br>test_flag_gap_pointers.py,<br>test_charter_report_quality.py,<br>test_charter_report_source_labels.py |
| <b>Citations</b>                              | test_citation_formatting.py,<br>test_external_citations.py                                                                                                                                             |
| <b>Scoring</b>                                | test_developer_score.py,<br>test_documentation_confidence_score.py, test_executive_ready.py                                                                                                            |
| <b>Data correctness</b>                       | test_facts_normalization.py,<br>test_data_quality_fixes.py,<br>test_finding_research.py                                                                                                                |
| <b>GST</b>                                    | test_gst_intake.py,<br>test_gst_portal.py,                                                                                                                                                             |

| Group                     | Files                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------------|
|                           | test_gst_compliance.py,<br>test_gst_compliance_check.py                                                      |
| <b>Intake / carryover</b> | test_cts_intake.py,<br>test_promoter_intake.py,<br>test_attach_rera_number.py,<br>test_promoter_portfolio.py |
| <b>Other</b>              | test_executive_briefing.py,<br>test_usage_tracking.py                                                        |

There is no tests/ package, no conf test.py, no CI configuration.

## 7. Runtime view — the end-to-end pipeline

```

=====
RERA SCRAPPER  --  END-TO-END PIPELINE
entry: main.py::main() | 11 stages
=====

[ENTRY]
+-----+
| main.py <REG_NO|name> |
| --gstln X | --pan Y   |
| --headed --token T    |
| --no-auto-auth         |
| --project-id N         |
| --output-dir D --verify|
+-----+
+-----+
| app.py (Streamlit)    |
| Run Scraper tab wraps |
| the SAME functions    |
| (no duplicated logic) |
+-----+
|
+-----+
|
v
pipeline_start_time = time.time() <-- fed to the Charter's version
log
|
=====
STAGE 1  --  RESOLVE
resolver.py
=====
is query ^P\d{11}$ ?
+-----+
| YES | NO
v     v
--project-id given?

```

```

search_projects(name)
|
| YES| NO|
|
| 0 hits +-->
exit(1)
|
| v
| resolve_project_id()
|
| 1 hit +-->
use it
| Playwright -> Drupal search
|
| N hits +-->
interactive
| read View Details href
|
| pick
| /public/project/view/(\d+)
|
| (no TTY ->
exit(2))
|
| *** never CLICKS the link:
| a click fires a JS confirm()
| and lands on the CAPTCHA page ***
|
+-----+
|
| v
| (project_id, detail_url, reg_no)
|
+-----+
=====
=====
STAGE 2 -- AUTH session_auth.py +
token_cache.py
=====
=====
|
| v
+-----+
| --token supplied? | --YES--> auth_source =
"explicit"
+-----+
| NO
|
| v
+-----+
| --no-auto-auth? | --YES--> auth_source = "none"
| (2 free categories
only)
+-----+
| NO
|
| v
+-----+
| token_cache.load_valid | --HIT--> auth_source =
"cached"
| MAX_AGE_MINUTES = 90 | (real TTL ~100 min;
| 10 min safety
margin)
+-----+
| MISS
|
| v
+-----+
| acquire_token_via_browser()
|

```



```

| headless=False ALWAYS |
| goto /public/project/view/{id} |
| fast path: read token once BEFORE prompting |
| poll sessionStorage['tokens'].accessToken |
|   every 2.0 s, up to 300 s |
|   status line every 30 s |
+-----+
| success | | timeout / closed |
|   |     | |   |         |
|   v     | |   v         |
| auth_source = | | CaptchaTimeoutError / |
| "fresh_browser" | | BrowserClosedError |
| token_cache.save() | | -> [WARN], continue |
|                   | | -> auth_source = "none" |
|                   | |                   |
|                   | | v                 |
+-----+

```

```

=====
=====

```

```

STAGE 3 -- ARCHIVE
run_archive.py

```

```

=====
=====

```

```

*** ORDER IS LOAD-BEARING ***
1. load_prior_research() <-- MUST be first: archiving
moves
research/deep_research.json
away
2. archive_previous_run() <-- shutil.move (not copy) of the
whole
output/<reg>/ tree into
output/_history/<reg>/<ts>/
collision -> _2, _3, ...
3. load_prior_manifest() <-- MUST be after: reads FROM the
archive
prior_documents_dir()
load_prior_complaint_orders_manifest()
prior_complaint_orders_dir()
|
deep_research.reset_usage_log()
(matters for app.py's long-lived process)
v

```

```

=====
=====

```

```

STAGE 4 -- SCRAPE
api_client.py

```

```

=====
=====

```

```

fetch_all_categories(project_id, raw_dir, token, errors_out=errors)

```

```

STEP A (serial, alone)          STEP B (parallel)
+-----+

```

```

+-----+
| projects | ThreadPoolExecutor(max_workers=8)
|
| getProjectGeneral |---->| spocs . professionals . partners
|
| DetailsByProjectId | | past_experiences . sro_details
|
| *** the ONLY endpoint | | documents . complaints . appeals
|
| marked "confirmed" | |
|
| the other 8 are | | each worker: its OWN
requests.Session |
| "observed" *** | | (Session thread-safety not
guaranteed) |
|
| carries userProfileId | | past_experiences needs
userProfileId |
| -> needed by | | from projects -> that is the ONE
real |
| past_experiences | | ordering dependency
|
+-----+

```

```

+-----+
| |
+-----+
v
_unwrap_envelope(): responseObject / data /
projects / content; a dict whose keys are a
SUBSET of _ENVELOPE_ONLY_KEYS returns {}
*** MahaRERA returns HTTP 200 with pure
bookkeeping, e.g. {"message":"ERROR",
"status":"0"} -- without this it renders
as a fake data row ***
v
write raw_dir/<category>.json
(on failure writes {"_error":...,"status_code":...}
so raw/ always has one file per category)
v
+-----+
| gated categories that returned 401/403? |
+-----+
| YES and auth_source != "none"
v
token_cache.invalidate() -> re-solve CAPTCHA
-> refetch ONLY the failed subset -> update
*** EXACTLY ONE RETRY. No backoff anywhere. ***
v
=====

```

```

=====
STAGE 5  --  DOCUMENTS                                [complaint orders:
NEVER FATAL]
=====
=====
download_documents()
download_complaint_orders()
+-----+
+-----+
| ThreadPoolExecutor(8) | | SERIAL, one Session
|
| _MAX_DOCUMENT_DOWNLOAD_ | | (complaint counts are
small;|
| WORKERS = 8 | | documents can be 100+)
|
|
| POST {"fileName","documentId"} | | reads orderDmsRefNo +
| to the DMS path with bearer| | orderFileName from BOTH
| *** NOT a static GET *** | | complaintDetails[] and
| | | | miscComplaintDetails[]
|
| Origin + Referer headers | |
|
| mimic the real detail page | | "has always been
present, | | it was simply never
|
| _sniff_and_save(): | | downloaded before"
|
| Content-Type is a LIE -- |
+-----+
| raw PDF bytes have been | |
| observed labelled | | manifest row schema:
| application/json. | | label . original_url .
saved_filename
| Trust JSON only if it | | status . method . document_id
| actually parses. | | source_filename [+
complaint_id,
|
| _dedupe_filename(): | | complaint_registration_no]
| two DIFFERENT documents | | status in {downloaded, reused,
failed,
| can share a filename | | failed (HTTP n), failed
(<exc>),
| -> " (2)", " (3)" before | | failed (couldn't resolve file
bytes)}

```

```

|   the extension   |
by |                 | REUSE: prior manifest indexed
| one threading.Lock guards | (document_id, source_filename)
for | seen_filenames + seen_urls | dms-post, by original_url for
+-----+ direct-url -> shutil.copy2,
status                                     "reused" (itself reusable ->
chains)
                                     v
=====
=====
STAGE 6  --  PROMOTER PORTFOLIO          [NEVER FATAL] own
browser
=====
=====
promoter name from partners.promoterDetails.promoterName
(promoterName is null on every sample seen)
|
v

+-----+
| build_promoter_portfolio()
|
| resolver.search_promoters() -> promoter's whole RERA portfolio
|
| cap: PROMOTER_PROJECT_LIMIT = 25 (never silent -> `truncated`)
|
| per project: projects / complaints / appeals / past_experiences
|
| geocode via OSM Nominatim, >= 1.1 s apart, per-run cache
|
| 5 km radius filter (haversine, r = 6371.0 km)
|
| a failed geocode is DROPPED, never treated as "0 km away"
|
| self-exclusion by the ENTRY's own reg-no / exact name
|
| (keying it on the fetch reg-no once destroyed real data
|
|   for single-project SPVs)
|
| emits 7 standing limitations[] + an 8th when truncated
|
+-----+
                                     v ->
output/<reg>/promoter/portfolio.json
=====

```

```

=====
STAGE 7  --  GST INTAKE                                [OPT-IN] [NEVER
FATAL] [HUMAN]
=====
=====
    only with --gstins or --pan (mutually exclusive)
    placed HERE, not after deep research, because it is the LAST step
    needing a human at the keyboard -- it sits beside the other browser
work
                                |
                                v
    format-validate BEFORE opening a browser (a typo must never
    cost a human a CAPTCHA solve)
                                v
PAN --> search_gstins_by_pan() [1 human CAPTCHA]
    --> for each GSTIN: fetch_gstin_filing_table() [1 human
CAPTCHA each,
    then ALL financial years walked in the same session]
    --> parse_filing_table() -> GSTR-1 / GSTR-3B periods only
    (CMP08, GSTR9/9C, GSTR1A deliberately NOT parsed: no
statutory
    due-date rule implemented, so scoring them would be a
guess)
    --> primary GSTIN = the one with the MOST periods
    (schema is single-GSTIN; QRMP due dates are state-
specific)
    --> output/<reg>/gst_filing_input.json
    *** an empty records:[] file is NEVER written -- the
file's
    existence means "GST data was supplied" ***
                                v
=====
=====
STAGE 8  --  DEEP RESEARCH                                [NEVER FATAL]
unattended
=====
=====
    deep_research.run_deep_research(reg_no, category_data, out,
prior_research)

+-----+
| | prior research < RESEARCH_REUSE_WINDOW_HOURS (24) ?
| |
| |     YES -> carry confirmed sources forward untouched (zero cost
| |
| |         for a block with no gaps); re-attempt ONLY open gaps
| |
| |     NO  -> full pass
| |
+-----+

```

```

                                v
_run_agentic_pass(label="research_generate")
  client.beta.messages.tool_runner(model="claude-sonnet-5",
    max_tokens=8000, tools=[{"type":"web_search_20260209",
                              "name":"web_search"}])
  runner is an ITERATOR -> one BetaMessage per turn -> list(runner)
  usage recorded across ALL turns (each web_search round-trip is
  separately billed; counting only the last would undercount)
                                v
three blocks: macro_market . micro_market . promoter_external
each: {summary, sections[{heading,body}], sources[{claim,url,
  publisher,accessed_date}], gaps[str]}
                                v
_verify_block(): EVERY cited source re-checked, BATCHED --
  _verify_claims_batch() checks up to BATCH_VERIFY_CHUNK_SIZE (10)
  sources per call, sharing one BATCH_VERIFY_MAX_SEARCHES (10)
search
  budget, instead of one independent call per source
  confirmed -> keep
  verification_error -> KEEP the source, append an honest gap
                                "(NOT independently re-verified this
pass ...)"
  unsupported/stale -> DROP the source into gaps
                                v
_resolve_gaps(): up to MAX_GAP_RETRY_ATTEMPTS = 2 rounds, each with
a
  DIFFERENT strategy (registry/official filing; related-party
angle) --
  _retry_gaps_batch() retries EVERY gap still open in a round in
ONE
  call sharing one BATCH_GAP_RETRY_MAX_SEARCHES (10) search budget,
  instead of one call per gap per attempt
  retry results are themselves re-verified (_verify_claims_batch)
before
  merging; a shared _VerificationBudget (MAX_RESEARCH_VERIFICATION_
CALLS = 15) still caps the total batched-call count across all
three
  blocks combined, as a backstop for an unusually large block/gap
count
  unresolved -> "(retried N alternate approach(es), not confirmed)"
  *** never fabricated away just to make the list shorter ***
                                v
  EVERY call in this stage funnels through _run_agentic_pass, which
  refuses to even START a call once this run's total spend (across
  deep research AND Charter generation) is >= PIPELINE_COST_CAP_USD
  ($6.00) -- raises CostCapExceeded, never retried, degrades this
  stage exactly like a missing ANTHROPIC_API_KEY would
                                v
                                output/<reg>/research/deep_research.json
                                stamped _generated_at + _reused_prior

```

```

v
=====
=====
STAGE 9  --  COMPANY CHARTER                                [NEVER
FATAL]
=====
=====
                                see section 9 for the full internal view
                                company_charter.run_company_charter(...)
                                |
                                v

output/company_charters/Company_Charter_<Name>_<REG>_Internal.pdf
                                ..._External.pdf
                                ....facts.json

Company_Charter_<REG>_claude_md_review.json
                                v
=====
=====
STAGE 10  --  RERA SUMMARY PDF
report.py
=====
=====
    build_pdf() -- ReportLab, no native deps
    cover -> Project Details -> Company Charter Highlights -> Promoter
Profile
    -> Market Research -> remaining CATEGORY_ORDER categories
    -> output/<reg>/<REG_NO>_summary.pdf
    *** a DIFFERENT document from the Charter. Do not confuse them. ***
                                v
=====
=====
STAGE 11  --  RUN SUMMARY + USAGE LOG
main.py
=====
=====
    also persists documents_manifest.json + run_meta.json
    write_usage_log() -> output/<reg>/usage_summary.json
                        -> append one rollup line to
output/usage_log.jsonl
    prints: auth source, per-category counts, documents downloaded,
            promoter_profile, market_research, gst_filing_intake,
            company_charter, claude_api_usage (calls / tokens / $ by
label)
=====
=====

SIDE PATH  --  manual, opt-in, never auto-run

```

```

+-----
-+
| mahabhumipy -- CTS -> Maha Bhulekh Property Card
|
| Requires a human to drop output/<reg>/cts_lookup_input.json with
the |
| office/village pre-resolved (python cts_resolve.py
offices/villages/ |
| candidates/finalize), because the site's CAPTCHA has NO reusable
|
| session -- solving it blocks on a human on EVERY call.
|
+-----
-+

```

---

## 8. Stage-by-stage specification

### 8.1 Stage 1 — Resolve (resolver.py)

| Aspect    | Detail                                                                                  |
|-----------|-----------------------------------------------------------------------------------------|
| Input     | P\d{11} registration number, or free-text project name                                  |
| Output    | (project_id, detail_url, reg_no)                                                        |
| Transport | Playwright Chromium, viewport 1920×1080, headless = not --headed                        |
| Endpoint  | https://maharera.maharashtra.gov.in/ —<br>Drupal, <b>no login, no CAPTCHA</b>           |
| Timeout   | SEARCH_TIMEOUT_MS = 30000; fixed 300 ms settle after a tab click                        |
| Retries   | None                                                                                    |
| Failure   | Any exception waiting for the first result link<br>– return [], treated as “no results” |

#### Key design decisions.

1. **Read the href, never click it.** Clicking “View Details” fires a JS confirm() and lands on the CAPTCHA-gated page. The href already contains /public/project/view/<id>, so id resolution is 100 % unauthenticated.
2. **Bounded ancestor walk.** \_CARD\_ANCESTOR\_WALK\_JS climbs at most 8 parent levels looking for innerText matching /P\d{11}/, then falls back to closest('div'). Bounded so it cannot run away to document.body.
3. **Selector scoping.** MahaRERA reuses id="edit-submit" across four hidden per-tab forms; the Projects submit is therefore scoped as #projects-form



input[value='Search'] and the Promoters submit is the ordinal #edit-submit--3.

4. **raw\_text is always retained** on ProjectCandidate so nothing is silently lost when a positional regex misses.
5. **Positional parsing is brittle by admission** — reg-no line, project name on the next line, promoter on the line after.
6. search\_promoters() returns a promoter's entire registered portfolio in the same card format — this is what makes Stage 6 possible.

## 8.2 Stage 2 — Auth (session\_auth.py, token\_cache.py)

The four auth\_source values are explicit, cached, fresh\_browser, none, and **the pipeline runs to completion in all four** — the last simply fetches only projects and complaints.

### TOKEN LIFECYCLE

-----

--token X -----> [explicit] no I/O

```
.guest_token_cache.json
{"token": "...", "saved_at": "<ISO>"}
|
| minutes_left() = max(0, 90 - age_minutes)
v
```

> 0 ? ---YES---> [cached]

|  
NO

v  
acquire\_token\_via\_browser()  
| visible Chromium, project detail page  
| fast path: read token BEFORE prompting  
| poll every 2.0 s, budget 300 s  
v

```
sessionStorage.getItem('tokens') -> JSON.parse -> .accessToken
|
+---success--> token_cache.save() --> [fresh_browser]
|
+---timeout/closed--> [WARN] -----> [none]
```

**Notes.** - MAX\_AGE\_MINUTES = 90 against a real token lifetime of roughly 100 minutes — a deliberate 10-minute safety margin so a token cannot expire mid-run. - Freshness is measured against local wall-clock saved\_at, **not** a decoded JWT exp. Clock changes skew it. - The token is stored **plaintext with default file permissions**. Mitigated by it being a short-lived guest token for public data (\$22). - elapsed in the poll loop is accumulated arithmetically, not wall-clock, so token-read latency is not charged against the 300 s budget; real wall time can exceed it. - A corrupt cache file is silently treated as absent (JSONDecodeError — None). There is no file locking or atomic write. - **Nothing solves the CAPTCHA**. Stated twice in the module docstring; an explicit design and ethics boundary.

### 8.3 Stage 3 — Archive (`run_archive.py`)

The call-order contract is load-bearing and is documented in the module docstring:

```
load_prior_research()    -- BEFORE archiving (archiving moves the
file away)
|
archive_previous_run()   -- shutil.MOVE, so no disk doubling and
|                          output/<reg>/ is guaranteed empty
afterwards
  v
load_prior_manifest()    -- AFTER archiving (reads FROM the archive)
prior_documents_dir()
load_prior_complaint_orders_manifest()
prior_complaint_orders_dir()
```

Charters live *outside* `output/<reg>/`, in `output/company_charters/`, so they were historically never archived — only overwritten. `archive_previous_charter()` closes that gap, and is what makes field-level change detection (`_diff_mortgage_lender`) possible. It matches by `<reg_no> suffix`, not exact filename, because the project-name segment can change between runs when a RERA record is renamed.

Timestamp format `%Y%m%d_%H%M%S`, with `_2`, `_3`... suffixes on same-second collision. Not safe against two concurrent runs of the same `reg_no` — the collision check is not atomic.

### 8.4 Stage 4 — Scrape (`api_client.py`)

The nine category endpoints, and their trust status as recorded in `config.CATEGORY_ENDPOINTS`:

| Category      | Path                                                                            | Status           | Auth  |
|---------------|---------------------------------------------------------------------------------|------------------|-------|
| projects      | .../<br>projectregistart<br>ion/<br>getProjectGenera<br>lDetailsByProjec<br>tId | <b>confirmed</b> | none  |
| complaints    | .../<br>projectregistart<br>ion/<br>getComplaintDeta<br>ilsByProjectId          | observed         | none  |
| professionals | .../<br>projectregistart<br>ion/<br>getProjectProfes<br>sionalByType            | observed         | token |
| partners      | .../<br>projectregistart<br>ion/                                                | observed         | token |

| Category         | Path                                                                                  | Status   | Auth  |
|------------------|---------------------------------------------------------------------------------------|----------|-------|
|                  | getProjectAndAssociatedPromoterDetails                                                |          |       |
| spocs            | .../<br>projectregistration/<br>getProjectSpocMapping                                 | observed | token |
| sro_details      | .../<br>projectregistration/<br>getProjectSroDetails                                  | observed | token |
| documents        | .../<br>projectregistration/<br>getUploadedDocuments                                  | observed | token |
| past_experiences | /api/maha-rera-promoter-management-service/<br>promoter/<br>getPromoterPastExpProject | observed | token |
| appeals          | /api/maha-rera-appeal-service/<br>reatapeal/<br>getAppealDetails<br>PublicView        | observed | token |

The misspelling projectregistration is **MahaRERA's own** and must be preserved verbatim.

**confirmed vs observed is a first-class, machine-readable concept.** confirmed means the exact payload/response shape was individually re-verified. observed means the endpoint name was taken from real successful traffic on the site's own page but was not re-verified. The distinction is surfaced in `fetch_all_categories`' warning lines, in the run summary, in `discover.verify_endpoints`' report table, and on the summary PDF cover as a **Data reliability note**.

#### Five engineering accommodations worth citing:

1. **Envelope normalisation** — MahaRERA returns HTTP 200 with pure bookkeeping (`{"message": "ERROR", "status": "0", "responseObject": null}`). `_ENVELOPE_ONLY_KEYS` subset detection collapses these to `{}` so they cannot render as a fake “message/status” data row.

2. **The past\_experiences 400** — that endpoint keys off userProfileId, not projectId, and rejects the generic body with HTTP 400. \_past\_experiences\_body reuses the id from the already-fetched projects response, or re-fetches it (cheap, no auth). This is the pipeline's one true ordering dependency.
3. **Documents are POST-downloaded** to the DMS service with a bearer token, not GET from a static URL. Recovered from observed working traffic.
4. **Content-Type is a lie** — raw PDF bytes have been observed labelled application/json. \_sniff\_and\_save trusts JSON only if the body actually parses, follows one level of nested url/downloadUrl/signedUrl/data, and otherwise streams raw bytes in 8192-byte chunks. An unrecognised parsed JSON returns False and writes nothing rather than a bogus file.
5. **Filename collisions are real** — two genuinely different documents (different document\_id) can share a filename. \_dedupe\_filename appends (2), (3) before the extension.

## 8.5 Stage 5-7

Covered in the diagram in §7. Two asymmetries worth recording explicitly:

- **Document download is parallel (8 workers); complaint-order download is serial.** Deliberate: complaint counts are small, document counts can exceed 100.
- **GST intake sits between the portfolio build and deep research**, not after it, because deep research runs unattended for minutes while GST needs a human at the keyboard. All human-attended browser work is contiguous.

## 8.6 Stage 10 — RERA summary PDF (report.py)

ReportLab only — deliberately no native dependencies, “deploys cleanly if this is ever hosted as a service”. A4, 2 cm margins.

Section order:

```

1  Cover ----- title, reg no, internal project id, project
name,
                                promoter, generated timestamp, "Data session:"
label,
                                + Data reliability note (every non-"confirmed"
category)
2  Project Details ----- from `projects`
3  Company Charter Highlights ----- Documentation Confidence
Score
    (from charter_facts only,          Credit Rating (ICRA +
Infomerics)
    nothing recomputed or refetched)  Insolvency Check (IBBI)
                                        Appeal-Level Judgments Found
                                        Mortgage Lender
4  Promoter Profile ----- totals, limitations[], per-
project
                                table (FLAGGED when
```

```

is_lapsed),
                                Promoter External Profile
research
  5  Market Research ----- Macro, then Micro (Locality)
  6  Remaining CATEGORY_ORDER ----- SPOCs, Professionals,
Partners,
                                Past Experiences, SR0
Details,
                                Documents, Complaints,
Appeals

Defensive details: _esc ( ) XML-escapes every value (a bare & aborts a ReportLab build);
_LONG_VALUE_THRESHOLD = 300 pulls long values out of fixed-width cells to avoid
LayoutError; _MAX_TABLE_COLS = 6. Footer on every page: MahaRERA report --
<reg_no> / Page <n>.

```

---

## 9. The Company Charter subsystem

`company_charter.py` is 10,182 lines, 226 module-level functions and one class. It is effectively **six layers in a single file**. This section decomposes it.

### 9.1 Internal architecture

```

+=====
=====+
|               company_charter.py  --  INTERNAL LAYERS
|
+=====
=====+

  (1) RULES / PROMPT ASSEMBLY                                lines 106-
163

+-----+
--+
| _read_rules_section(marker) <-- regex over rules.md, raises on
|
|                               missing or empty section
|
| _coding_time_notes()      Section A  -- NEVER transmitted
|
| _common_content_rules()   Section B  -- every content call
|
| _external_citation_rule() Section C  -- external calls only
|
| _charter_system_blocks(external=, extra=)
|
|     -> [B as cache_control ephemeral prefix] (+ C) (+ extra)
|

```

```
+-----+
--+
(2) INGESTION + FACTS PASS | lines 166-
716
```

```

+-----+
+--+
| | _select_documents_for_extraction -> (extracted_text_by_label,
| |                                     all_labels_with_status)
| |
| | _extract_document_text  -- PyMuPDF, falls back to Tesseract OCR;
| |                           a missing Tesseract degrades to a
| |                           "[OCR unavailable]" marker, never fatal
| |
| | summarize_complaint_outcomes . summarize_professionals
| |
| | _normalise_entity_name . _is_llp . _is_partnership . _role_word
| |
| |
| | _CHARTER_FACTS_SCHEMA (JSON Schema) -> _CHARTER_JSON_SHAPE
| |
| |                                     -> embedded in _SYSTEM_PROMPT
| |
| | _run_charter_pass()  label="charter_pass"
|

```

```
+-----+
--+
      |
(3) EXTERNAL ENRICHMENT (5-way parallel)          lines 1131-
3078
```

|   |                 |                                          |
|---|-----------------|------------------------------------------|
| + | ---             | +                                        |
|   | credit rating   | ICRA + Infomercials, both, side by side  |
|   | insolvency      | IBBI public search -> _classify_ibbi_hit |
|   | company profile | _run_mca_profile_chain (see 9.3)         |
|   | group companies | ZaubaCorp ONLY, corroborated by          |
|   |                 | InstaFinancials directorship counts      |

```

| judgments          MahaRERA /orders-judgements + retry
|
|                   -> cross_reference_appeals
|
|                   -> _save_appeal_judgment_pdfs
|
| CTS land record    run_cts_land_lookup [opt-in, human]
|
| GST compliance     run_gst_compliance_check [opt-in]
|
| maps distances     _refine_distances_with_maps [env-gated]
|

```

```

+-----+
--+
```

```

(4) EDITORIAL / NORMALISATION | lines 4485-5310

```

```

+-----+
--+
```

```

| _normalize_misfiled_facts -> _relocate_merge_orders
|
|
|_point_rera_landowner_at_identity_table |
|                                     (NOT reversed -- a relocation loses
|                                     nothing, the corrected record is
|                                     the better one)
|
| run_editorial_passes -> 3 model passes, cached on `facts`
|
| run_finding_research -> per-finding deep research, capped at 8
|
| _scrub_clean_checks / _restore_clean_checks (REVERSIBLE)
|
| _sanitize_process_gaps / _pre_sanitize_gaps (REVERSIBLE)
|

```

```

+-----+
--+
```

```

(5) SCORING | lines 7002-8263

```

```

+-----+
--+
```

```

| _classify_flags -> {imminent[], structural[], monitor[]}
|

```

```

| | stable 1-based gap_number per gaps[i]
| | _compute_developer_score          3 buckets, 9 sub-metrics, NO
renorm |
| | _compute_documentation_confidence_score 8 criteria, RENORMALISES
| | verify_cross_corroboration (runs LAST so nothing looks single-
| | sourced mid-assembly)
| | _record_source_hits_and_promote -> source_trust_registry.json
|

```

```

+-----+
--+

```

```

|
(6) RENDERING -- _fill_template, ~733 lines lines
5311-6042

```

```

+-----+
--+

```

```

| | INTERNAL first (real facts, computes the persisted scores)
| | EXTERNAL second (_externalized_facts_copy -- deep copy)
| | see section 18
|

```

```

+-----+
--+

```

```

|
(7) GATES lines 4191-4332, 4943-
5155

```

```

+-----+
--+

```

```

| | _preflight_rules -> _verify_external_document_quality
| | -> run_claude_md_document_review [BLOCKS
PDF] |
| | see section 13
|

```

```

+-----+
--+

```

## 9.2 run\_company\_charter — the 32-step orchestration

```

def run_company_charter(
    reg_no, category_data, documents_manifest, documents_dir,
    research_data=None, output_dir=config.OUTPUT_ROOT,

```



```

    complaint_orders_manifest=None, complaint_orders_dir=None,
    reviews=None, review_source_label=None, promoter_portfolio=None,
    pre_built_facts=None, pipeline_start_time=None,
) -> tuple[str, dict]          # (internal PDF path or .docx fallback,
facts)

    01 _charter_start_time = time.time(); assert TEMPLATE_PATH exists
    02 _select_documents_for_extraction() -> extracted_docs,
doc_library_status
    03 FACTS: pre_built_facts, or _run_charter_pass(user_prompt)
        user_prompt = reg_no + projects JSON + partners JSON
        + extracted doc text (truncated to
_MAX_TOTAL_DOC_CHARS)
        + full document-library name list
        + research_context from
deep_research.RESEARCH_KEYS
    04 _verify_material_claims(facts)
label="material_claim_verify"
    05 _check_document_grounding(facts, ...)
label="document_grounding_verify"
                                (+ optional Grok
fallback)
    06 facts["document_library"] = doc_library_status    <- CODE,
overwrites model
    07 facts["professional_team"] =
summarize_professionals(category_data)
    08 facts["promoter_portfolio"] = promoter_portfolio    <- CODE
    09 facts["complaint_outcomes_summary"] =
summarize_complaint_outcomes(...)
    10 extract CIN / LLPIN; derive identifier_label; project name for
judgments
    11 _load_promoter_carryover() -- reuse a prior CIN-only intake if
attached
    12 +----- 5-WAY PARALLEL
-----+
        | ThreadPoolExecutor(max_workers=5)
    |
    | | _safe_credit_rating    _safe_ibbi_check
_safe_company_profile |
    | | _safe_group_companies _safe_judgments_search
    |
    | (first four SKIPPED entirely when a promoter carryover was
used)|

+-----+
    13 fold results in sequentially; append one `sources` entry per hit
        roster_conflicts -> facts["gaps"]    (renders in BOTH
variants)
    14 run_review_authenticity_triage(reviews, facts)    [if
reviews.json present]

```

```

15 run_cts_land_lookup(facts, reg_no, output_dir)      [opt-in]
16 run_gst_compliance_check(facts, reg_no, output_dir) [opt-in]
17 _refine_distances_with_maps(facts, origin)
[COMPANY_CHARTER_USE_MAPS_SCRAPES=1]
18 verify_cross_corroboration(facts)                  <- deliberately LAST
19 mkdir output/company_charters/
20 archive_previous_charter -> load_prior_charter_facts
   -> _diff_mortgage_lender ->
facts["mortgage_lender_history_note"]
21 facts["source_promotion_notes"] =
_record_source_hits_and_promote(...)
22 _normalize_misfiled_facts(facts)
23 run_editorial_passes(facts)                        [try/except -> deterministic
fallback]
24 run_finding_research(facts)                       [try/except -> original text
kept]
25 read_deep_research.usage_summary()["total"]; compute elapsed
26 _fill_template(..., doc_variant="internal", elapsed=, cost=,
calls=)
27 _fill_template(..., doc_variant="external")
28 run_claude_md_document_review(...) *** MAY RAISE
CharterComplianceError
                                                                    BEFORE ANY PDF EXISTS
***
29 _convert_docx_to_pdf(internal) ; _convert_docx_to_pdf(external)
30 _restore_clean_checks(...) ; restore facts["gaps"]
31 write Company_Charter_<Name>_<REG>.facts.json
32 return (internal_pdf_path or out_path, facts)

```

**Internal renders first, and this is not arbitrary.** Internal renders on the real facts dict and computes the scores that get persisted; External renders from a deep copy. Reversing the order would persist externalised values.

### 9.3 The MCA-mirror redundancy chain

This is the only place in the system where three independent sources are **cross-checked against each other** rather than merely falling back on failure.

```
_run_mca_profile_chain(cin, promoter_name)
```

```

+-----+
|
|
|   Zaubacorp -----+
|   (CIN redirect trick) |
|
|
|
|   Tofler -----+----> _merge_director_rosters()
|

```

```

|   (Playwright: type   |
|   the name in a real  |   +-- rosters MATCH
|   browser, verify CIN)|   -> merge silently
|
|   InstaFinancials -----+   +-- rosters DISAGREE
|   (ASP.NET web service,|   -> roster_conflicts
|   CIN search direct) |   -> facts["gaps"]
|
|   variants |   -> renders in BOTH

```

+-----+

REAL CASE HIT: Zaubacorp reported director "Cooper";  
 InstaFinancials  
 reported "Sahaya", for the same CIN. BOTH were surfaced. Neither  
 was  
 silently picked. This is the architectural pattern the whole system  
 is built around: two contradictory facts beat one invented one.

Group companies come from Zaubacorp ONLY -- Tofler and  
 InstaFinancials  
 do not expose that data freely -- and are corroborated against  
 InstaFinancials directorship counts (\_corroborate\_group\_companies).

#### 9.4 Source-trust registry (cross-run, persistent)

```

facts["sources"] -> _record_source_hits_and_promote(facts,
reg_no)
|
|   _extract_open_web_domain() -- excludes MahaRERA, MCA
mirrors,
|
|   rating agencies, IBBI, Maps
|   (already trusted by design)
v
tally each open-web domain's hit count across ALL runs
(99acres, NoBroker, press, social, Wikipedia, ...)
|
|   5th DISTINCT project -> AUTO-PROMOTE
v
source_trust_registry.json (repository root, accumulates forever)
|
v
"Source Trust Registry Updates (this run)" -> INTERNAL document
only

```

This is the system's only piece of cross-engagement learned state. It is committed to the repository (1,565 bytes at time of writing), which makes promotions reviewable in version control.

---

## 10. Data architecture

### 10.1 Filesystem layout

```

RERA_Scrapper/
├── *.py                                29 application modules
(flat)
├── test_*.py                          28 test modules (flat)
├── CLAUDE.md  rules.md  guardrails.md  README.md
CHARTER_RESHAPE_SPEC.md
├── requirements.txt                    the ONLY dependency
manifest
├── source_trust_registry.json          *** cross-run learned
state ***
├── .guest_token_cache.json             *** short-lived secret ***
├── RERA_Executive_Tracker.xlsx          operator-facing tracker
├── tessdata/
│   ├── eng.traineddata                 4.1 MB
│   └── mar.traineddata                 3.2 MB    (Marathi, for
Maha Bhulekh)
├── .claude/
│   ├── launch.json  settings.local.json
│   └── skills/maharera-report/SKILL.md
├── docs/                               11 .docx design artefacts,
no source
├── output/                             *** GITIGNORED ***
│   ├── usage_log.jsonl                 one rollup line per run,
forever
│   ├── _history/<REG_NO>/<YYYYmmdd_HHMMSS>/  archived prior runs
│   ├── pending/<CIN | district_village_cts>/  pre-RERA intake cases
│   │   ├── promoter_profile.json
│   │   ├── land_record.json
│   │   └── property_card_screenshot.png
│   ├── <REG_NO>/
│   │   └── raw/                         9 category JSONs, one per
category
│   │   ├── projects.json  partners.json  professionals.json
│   │   ├── spocs.json  sro_details.json  past_experiences.json
│   │   ├── documents.json  complaints.json  appeals.json
│   │   └── documents/      downloaded project
documents (PDF)
│   ├── complaint_orders/      complaint order PDFs
│   └── appeal_judgments/      appeal judgment PDFs

```

```

|   |   | promoter/portfolio.json
|   |   | research/deep_research.json
complaint_orders_manifest.json |   |   | documents_manifest.json
|   |   | run_meta.json           usage_summary.json
|   |   | gst_portal_raw/         per-lookup .png + .json
|   |   | gst_filing_input.json   <- feeds
run_gst_compliance_check
|   |   | cts_office_candidates.json
cts_village_candidates.json |   |   | cts_number_candidates.json   cts_lookup_input.json
|   |   | *_carryover.json         promoter_profile /
land_record
|   |   | reviews.json             (human-supplied, optional)
|   |   | <REG_NO>_summary.pdf      <- Stage 10 deliverable
|   |   | company_charters/        *** THE PRIMARY
DELIVERABLE DIR ***
|   |   | Company_Charter_TEMPLATE_WebSourced.docx <- the .docx
template
|   |   | Company_Charter_<Name>_<REG>_Internal.docx / .pdf
|   |   | Company_Charter_<Name>_<REG>_External.docx / .pdf
|   |   | Company_Charter_<Name>_<REG>.facts.json   <- THE FULL
RECORD
|   |   | Company_Charter_<REG>_claude_md_review.json
|   |   | _history/<REG_NO>/<timestamp>/
|   |   | _verify/                 cheap no-API render loop
output
|   |   | CHARTER RESHAPE CHANGE LOG.md

```

**rules.md \$A: Charter files are only ever saved to output/company charters/. Never to a Desktop folder or any other location.**

**The template lives under gitignored output/.** It must be backed up with a timestamped copy before any change touching its structure.

## 10.2 Artefact lifecycle

```

RUN N-1                                RUN N
-----                                -----
output/<reg>/ -----> output/_history/<reg>/<ts>/
    |                                |
    | documents_manifest.json        | prior_manifest
    | + documents/                   | + prior_documents_dir
    |                                |
    |                                | v
source_filename) reuse by (document_id,
    |                                |
    |                                | -> shutil.copy2 -> status "reused"
chains)                                | ("reused" is itself reusable ->
    |                                |
research/deep research.json ---> prior research

```

```

(24)? | within RESEARCH_REUSE_WINDOW_HOURS
      | -> carry confirmed sources
forward, | re-attempt ONLY open gaps
        |
        company_charters/*.facts.json ->
output/company_charters/_history/...
      | -> _diff_mortgage_lender vs
current |
        | ->
facts["mortgage_lender_history_note"]
      v
      source_trust_registry.json ----> accumulates ACROSS ALL PROJECTS,
forever
      output/usage_log.jsonl ----> append-only cost ledger

```

### 10.3 The two records that deliberately differ

```

facts (in memory)
  |
  +--> _scrub_clean_checks()    stash _pre_scrub_narrative
  +--> _sanitize_process_gaps() stash _pre_sanitize_gaps
  |
  v
  RENDERED DOCUMENT ..... the REDACTED view
  |                         (clean checks deleted, process failures
  |                         sanitised, file paths and module names
gone)
  v
  _restore_clean_checks() + restore facts["gaps"]
  |
  v
  .facts.json ..... the COMPLETE record
                        (what was checked stays in the record
even
                        when it leaves the page)

```

**Consequence for consumers:** anything reading `.facts.json` — `charter_report.py`, `executive_briefing.py`, `finalize_report.py`, `cts_resolve.py` — gets the **unscrubbed** text. Without the restore step, every run would quietly hollow out the file the next run reads.

---

## 11. The facts data model

`_CHARTER_FACTS_SCHEMA` is a JSON Schema dict serialised into `_SYSTEM_PROMPT`. It has **28 required top-level keys** (everything except `developer_track_record`, optional by design).

### 11.1 Building blocks

```
_FIELD_WITH_SOURCE = {"type": "object",  
  "properties": {"value": {"type": "string"}, "source": {"type":  
    "string"}},  
  "required": ["value", "source"]}  
_PLAIN_FIELD = {"type": "string"}  
_INT_FIELD = {"type": "integer"}
```

\_FIELD\_WITH\_SOURCE is the schema-level expression of goal **G1**: a sourced field cannot exist without its source.

### 11.2 Model-authored keys

| Key                                                                      | Shape                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| methodology_note,<br>executive_summary                                   | plain                                                                                                                                                                                                                                         |
| land_identification                                                      | 7 × sourced: survey_cts_plot_numbers,<br>village_locality,<br>mandal_taluka_district, pincode,<br>total_gross_area, area_affected,<br>net_area (+ land_assembly added at<br>runtime)                                                          |
| corporate_identity                                                       | 9 × sourced: promoter_name,<br>organization_type, cin_llpin,<br>registered_office_main,<br>registered_office_board_resolutio<br>n,<br>registered_office_planning_stage,<br>authorized_signatory,<br>partners_directors,<br>landowner_investor |
| address_discrepancy_note,<br>corporate_registry_cross_check              | plain                                                                                                                                                                                                                                         |
| litigation_status                                                        | sourced (top-level, not nested)                                                                                                                                                                                                               |
| location_coordinates_note                                                | plain                                                                                                                                                                                                                                         |
| neighbourhood                                                            | east, west, north, south                                                                                                                                                                                                                      |
| distances                                                                | array of {landmark, distance_time,<br>route_note}                                                                                                                                                                                             |
| connectivity                                                             | road, rail, metro, air                                                                                                                                                                                                                        |
| social_infrastructure,<br>fsi_governing_framework,<br>fsi_interpretation | plain                                                                                                                                                                                                                                         |
| fsi_metrics                                                              | net_land_area, approved_bua,<br>sanctioned_bua, mortgage_area,<br>implied_fsi + <b>mortgage_lender</b><br>(sourced)                                                                                                                           |

| Key                                                              | Shape                                                                                                                                                     |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules_statutory                                                  | governing_act,<br>planning_approval_sequence,<br>allotment_mechanics                                                                                      |
| rera_compliance                                                  | registration_summary,<br>collection_account,<br>escrow_subaccounts,<br>litigations_complaints_appeals,<br>statutory_declaration,<br>construction_progress |
| local_planning                                                   | authority_of_record, project_type,<br>professionals_of_record                                                                                             |
| micro_market_overview,<br>area_intelligence_trend<br>comparables | plain<br><br>array of {project, configuration,<br>pricing, source, distance_km} —<br>distance_km is a bare number                                         |
| rera_core_fields                                                 | 13 plain + total_complaints_count,<br>total_appeals_count, units_total,<br>units_sold (int),<br>completion_date_current,<br>completion_date_original      |
| developer_track_record                                           | years_in_industry (int) +<br>years_in_industry_basis — <b>omit<br/>entirely if unconfirmable</b>                                                          |
| unit_summary_note                                                | plain                                                                                                                                                     |
| blocks                                                           | array of {block_wing, floors,<br>config, units_counted, note}                                                                                             |
| documents_reviewed_note,<br>documents_absent_note                | plain                                                                                                                                                     |
| gaps                                                             | array of strings                                                                                                                                          |
| sources                                                          | array of {label, ref, topic,<br>published_date, accessed_date}                                                                                            |

**Enumerated topic values** — these drive both corroboration counting and automatic inline citation resolution:

land\_title·corporate\_identity·litigation·pricing·market\_trend·distance·  
project\_registration·legal\_documents·reputation·credit\_rating·  
insolvency\_status·company\_profile·group\_companies

### 11.3 Code-computed keys (never model-authored)

These are attached verbatim after the LLM pass and **overwrite** anything the model produced for the same key:



```
document_library·professional_team·promoter_portfolio·
complaint_outcomes_summary·credit_rating_check·ibbi_insolvency_check·
company_profile_check·group_companies_check·appeal_judgments_found·
review_authenticity_triage·cts_land_record_check·cts_mismatch_note·
gst_compliance_check·mortgage_lender_history_note·
source_promotion_notes·developer_score·documentation_confidence_score·
finding_research·claude_md_review·verification_stats
```

#### 11.4 Ephemeral keys (stripped or restored before persist)

```
_doc_variant·_citation_registry·_clean_check_verdicts·
_claim_source_matches·_flag_headlines·_editorial_passes·
_pre_scrub_narrative·_pre_sanitize_gaps
```

**Invariant, stated in three modules:** facts["developer\_score"] is an **output** of a prior run, never a valid **input**. charter\_report.py and executive\_briefing.py both recompute \_classify\_flags and \_compute\_developer\_score rather than trusting the persisted value.

---

## 12. Governance architecture — how rules.md is enforced

Rendering is pure code, so most of rules.md cannot be enforced by a model reading it. Four mechanisms, in execution order:

```
+=====
=====+
|               FOUR ENFORCEMENT MECHANISMS, IN ORDER OF EXECUTION
|
+=====
=====+

    (1) PREFLIGHT                                _preflight_rules(doc_variant)

+-----+
| The FIRST thing _fill_template does, before a paragraph is
written.|
|   a) every required section loads and is non-empty
|
|       (A + B always; C when external)
|
|   b) B -- and C when external -- contain NO em dash and NO "
-- "
|   c) A is loaded but NOT RETURNED, so preflight can never
become |
|       a route by which coding-time guidance reaches an API call
|
|   Failure -> RuntimeError. Nothing is written.
|
```

```

+-----+
|                                     |
| (2) PROMPT                         | _charter_system_blocks()
|                                     |
+-----+
|                                     |
| block | Section B -> a separate cache_control:{"type":"ephemeral"}
|       |           on EVERY content-generating or -checking call
|       |
|       | Section C -> appended after B, external-facing calls ONLY
|       |
|       | Section A -> NEVER
|       |
|       |
|       | Note: the External document is generated ENTIRELY BY CODE.
|       | Section C therefore reaches only the advisory citation JUDGE
|       | (_llm_verify_citation_completeness), never a call that writes
|       | External content. Any External behaviour must be implemented
|       | in code -- editing the rules alone will not move it.
|       |
|       |
+-----+
|                                     |
| (3) DETERMINISTIC PASSES          |
|                                     |
+-----+
| _normalize_misfiled_facts  -- corrections FIRST
|
| _scrub_clean_checks        -- scrub SECOND (order matters)
|
| _sanitize_process_gaps
|
| _remove_empty_section_headings . _remove_gap_rows
|
| _remove_fully_empty_rows . _consolidate_bullet_clauses
|
| _fix_bullet_capitalization . _expand_jargon_first_use
|
| _strip_inline_cin_din . _externalize_prose (external only)
|
| _center_all_table_cells (length-thresholded) . _apply_justify
|
| _apply_table_pagination (cantSplit + tblHeader)
|

```

|           |                                                          |
|-----------|----------------------------------------------------------|
| +-----+   |                                                          |
| (4) GATES | see section 13                                           |
| +-----+   |                                                          |
| save      | _verify_external_document_quality -> blocks the External |
|           | run_claude_md_document_review -> blocks the PDF          |
|           |                                                          |
| +-----+   |                                                          |

## 12.1 The core editorial rules (Section B), and where each is enforced

| Rule                                                  | Prompt | Deterministic pass                                        | Gate              |
|-------------------------------------------------------|--------|-----------------------------------------------------------|-------------------|
| A clean check produces no sentence                    | ✓      | _scrub_clean_checks /<br>_is_clean_check_clause           | _is_cited_absence |
| An empty section keeps its heading + "Nothing found." | ✓      | _remove_empty_section_headings<br>(preserves the heading) | review            |
| Findings first inside anything that survives          | ✓      | —                                                         | review            |
| Absence as evidence (3 carve-outs)                    | ✓      | _is_carved_out /<br>_REVIEW_CARVE_OUT_PATTERNS            | review            |
| A gap is not an absence                               | ✓      | _is_cited_absence returns False for Gap N.                | ✓                 |
| Deep research on every finding                        | ✓      | run_finding_research                                      | —                 |
| A live right is a finding                             | ✓      | _merged_mortgage_value                                    | —                 |
| Ruled-out items survive, in the right section         | ✓      | _relocate_merge_orders                                    | —                 |
| Say it once                                           | ✓      | _point_rera_landowner_at_identity_table                   | review            |
| Flags summarise, Gaps explain                         | ✓      | _flag_headline + (Gap N) pointers                         | review            |
| Internal keeps                                        | ✓      | _sanitize_process                                         | ✓                 |

| Rule                                            | Prompt | Deterministic pass                                                 | Gate                     |
|-------------------------------------------------|--------|--------------------------------------------------------------------|--------------------------|
| process failures,<br>External does not          |        | s_text,<br>_externalize_pro<br>se                                  |                          |
| No jargon, plain<br>language, keep key<br>terms | ✓      | _expand_jargon_f<br>irst_use                                       | review                   |
| Gloss raw status<br>strings                     | ✓      | —                                                                  | review                   |
| Bullets and grammar                             | ✓      | _consolidate_bul<br>let_clauses,<br>_fix_bullet_capi<br>talization | ✓ (numbering)            |
| CIN/DIN placement                               | ✓      | _strip_inline_ci<br>n_din                                          | carve-out in review      |
| Empty table/row<br>removal                      | ✓      | _remove_gap_rows<br>,<br>_remove_fully_em<br>pty_rows              | —                        |
| Numbered citations<br>[N] (Section C)           | ✓      | _register_citati<br>on,<br>_insert_marker_a<br>t_clause_end        | ✓ (parens,<br>numbering) |
| Marker placement at<br>clause end               | ✓      | _insert_marker_a<br>t_clause_end                                   | advisory judge           |
| Descriptive source<br>labels                    | ✓      | _external_source<br>_label,<br>_generic_one_lab<br>el              | ✓                        |

## 13. Guardrail architecture

### 13.1 The gate chain

facts assembled

|  
v

```
+-----+
| GATE 1  -- _preflight_rules(doc_variant)
|
| Stops: generation, BEFORE a paragraph is written
|
| Checks: rules.md missing . section marker broken . section empty
|
|           . B or C carrying an em dash or double-hyphen dash
```

```

|
|   Raises: RuntimeError
|
+-----+
|
|   v   render Internal .docx   (no external gate applies)
|   v   render External .docx
|   |
|   v
|
+-----+
|   GATE 2  --  _verify_external_document_quality(docx_path)
|
|   Stops: the External .docx SAVE
|
|   Re-opens the just-saved file. ELEVEN checks:
|
|
|   per paragraph (body AND every table cell, recursively):
|
|       1. " -- " hyphen-pair dash
|
|       2. em dash
|
|       3. leftover Internal Document Library status text
|
|           (_EXTERNAL_DOC_LIBRARY_LEFTOVER_RE)
|
|       4. *** a citation marker attached to an absence ***
|
|           (_is_cited_absence) -- the specific defect the whole
|
|           clean-check rule exists to prevent
|
|   per run:
|
|       5. italic placeholder styling (carve-out: the Standing Gap
|
|           paragraph is intentionally red C000000 + italic)
|
|       6. run colour outside _EXTERNAL_ALLOWED_RUN_COLORS
|
|   per Sources entry (body paragraphs matching ^\[ (\d+) \] \s+ \S ):
|
|       7. unbalanced citation parentheses
|
|       8. lost bullet numbering (pPr.numPr is None)
|

```

```

|   per table:
|
|   9. Weight column surviving in the Developer Score table
|
|   10. Weight column surviving in Documentation Confidence
|
|   numbering part:
|
|   11. abstractNum id=2 level 0 missing its hanging-indent fix
|
|
|   Raises: RuntimeError listing every violation
|
+-----+
|
|   v
|
+-----+
**| GATE 3  --  run_claude_md_document_review(...)  *** BLOCKS PDF
|
| Stops: THE PDF -- the actual deliverable
|
| Per variant: _rendered_document_text(path, limit=60000)
|
|   -> deep_research.review_document_against_rules(text, rules, v)
|
|   rules = Section B (internal) | Section B + C (external)
|
|   *** Section A is NEVER sent; test_claude_md_doc_review.py
|
|       asserts this, and now asserts the marker is PRESENT before
|
|       asserting it is ABSENT, after the test was found passing
|
|       vacuously ***
|
|
|   TWO ways to fail:
|
|   (a) a VERIFIED violation
|
|   (b) a review that could not run AT ALL
|
|       -- unverified is not the same as clean
|
|

```

```

|   Writes: Company_Charter_<REG>_claude_md_review.json
|
|   Raises: CharterComplianceError (the module's only class)
|
|   Override: CHARTER_ALLOW_UNCHECKED=1|true|yes -> advisory
|
|               "a decision to ship an unchecked document,
|
|               not a convenience"
|

```

```

+-----+
|
|   v
|   _convert_docx_to_pdf(internal) ; _convert_docx_to_pdf(external)
|   v
|   THE DELIVERABLE
|

```

### 13.2 Why blocking on a model's judgement is safe

```

model returns violations[]
|
|   v
|   _verified_violations(violations, document_text)
|
+-----+
|   normalise whitespace + case
|
|
|   quote length >= 12 normalised chars ?    --NO--> UNVERIFIABLE
|
|       | YES                                (cannot block)
|       v
|   quote literally present in the document ? --NO--> UNVERIFIABLE
|
|       | YES
|       v
|   _is_carved_out(raw) ? -----YES-----> UNVERIFIABLE
|
|       | NO                                discarded_reason=
|       v                                "explicitly permitted
|   VERIFIED -> BLOCKS                    by rules.md"
|

```

+-----+

The model's role is reduced from "decide whether this complies" to "POINT AT the offending text", and the pointing is then checked mechanically. An invented or paraphrased quote is logged as unverifiable and CANNOT stop a good document. Quotes under 12 characters are not trusted -- they match almost anything by coincidence.

```
_REVIEW_CARVE_OUT_PATTERNS:
^\s*nothing found\.\?\s*$
^\s*gap \d+\.
\b\d+\s+complaints?\s*/\s*\d+\s+appeals?\b
\b(?:no completion extension|0 complaints, 0 appeals)\b
CIN/DIN identity-table id pattern
^\s*N/?A\b
```

### 13.3 Never-fatal wrappers

| Wrapped in         | Stages                                                                                      |
|--------------------|---------------------------------------------------------------------------------------------|
| main.py            | GST intake, complaint-order download, promoter portfolio, deep research, Charter generation |
| company_charter.py | the editorial passes, per-finding research, the document review (when non-strict)           |

**The strongest case is run\_finding\_research:** a researcher that raises costs **only its own finding**, and no finding is ever deleted whatever the researcher returns — including on an empty or malformed reply. *Losing a finding to an expired token would be worse than never running the stage.*

main.\_run\_gst\_intake\_step exists as its own named function **purely so that contract is testable** rather than asserted in a comment.

### 13.4 Fallbacks — a model can only match or improve

|                                                |     |                                                                           |
|------------------------------------------------|-----|---------------------------------------------------------------------------|
| MODEL PASS                                     |     | DETERMINISTIC PREDECESSOR                                                 |
| -----                                          |     | -----                                                                     |
| classify_clean_checks<br>("clean_check_judge") | --> | risk nouns + field allow-list<br>_is_clean_check_clause                   |
| match_claims_to_sources<br>("citation_match")  | --> | keyword topic table<br>_clause_topic_citation<br>+ _CLAUSE_TOPIC_PATTERNS |
| write_flag_headlines<br>("flag_headline")      | --> | first sentence<br>_split_into_bullet_clauses[0]                           |

run\_editorial\_passes precomputes all three ONCE per document set and



caches them on `facts`, KEYED BY CLAUSE TEXT.

\*\*\* A lookup miss is indistinguishable from "no model ran". \*\*\*

Therefore a failed, partial or malformed reply produces EXACTLY the deterministic output. Malformed replies are discarded rather than trusted: out-of-range ids, unrecognised verdict kinds and null matches are all dropped.

### 13.5 Bounds

| Bound                                         | Value  | Purpose                                                                                                                                                     |
|-----------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| deep_research.MAX_FINDING_RESEARCH_CALLS      | 8      | Findings beyond the cap keep their original text rather than being dropped                                                                                  |
| deep_research.MAX_GAP_RETRY_ATTEMPTS          | 2      | One retry round per strategy, two strategies; every gap still open in a round is retried together, in one batched call                                      |
| deep_research.MAX_RESEARCH_VERIFICATION_CALLS | 15     | Shared<br>_VerificationBudget across all three research blocks, bounding the total number of batched verify/gap-retry calls in one run_deep_research() call |
| deep_research.BATCH_VERIFY_CHUNK_SIZE         | 10     | Sources per<br>_verify_claims_batch call; a block with 25 sources takes 3 calls, not 25                                                                     |
| deep_research.BATCH_VERIFY_MAX_SEARCHES       | 10     | Shared web-search budget for one<br>_verify_claims_batch call, across every claim in that chunk                                                             |
| deep_research.BATCH_GAP_RETRY_MAX_SEARCHES    | 10     | Shared web-search budget for one<br>_retry_gaps_batch call, across every gap still open in that round                                                       |
| deep_research.PIPELINE_COST_CAP_USD           | \$6.00 | Hard ceiling on total run spend (deep research + Charter generation combined), enforced in _run_agentic_pass;                                               |

| Bound                                       | Value        | Purpose                                                                        |
|---------------------------------------------|--------------|--------------------------------------------------------------------------------|
|                                             |              | refuses a call that would start after the cap, never retried (CostCapExceeded) |
| company_charter._MIN_FINDING_LENGTH         | 80           | A fragment is not a finding                                                    |
| review input cap                            | 60,000 chars | <code>_rendered_document_text(path, limit=60000)</code>                        |
| <code>_MAX_DOCUMENT_DOWNLOAD_WORKERS</code> | 8            | Download pool                                                                  |
| PROMOTER_PROJECT_LIMIT                      | 25           | Portfolio fan-out; never silent (truncated field)                              |

`deep_research.MAX_RESEARCH_VERIFICATION_CALLS` used to bound a fan-out that scaled one API call per source and per gap; as of 2026-08-13 `_verify_claims_batch` and `_retry_gaps_batch` fixed the root cause by checking/retrying many at once per call, so this budget is now a backstop for a block or gap count large enough to need several chunks, rather than the primary defence.

`_field_fingerprint` and `_already_researched` stop per-finding research

**recompounding:** an enriched finding is multi-sentence, so it re-splits into more clauses each pass and would be researched again. Any edit changes the fingerprint, so skipping is only ever an optimisation over identical input.

### 13.6 Reversibility

```

_scrub_clean_checks    <---->  _restore_clean_checks
_sanitize_process_gaps <---->  restore from _pre_sanitize_gaps
|
|    Both stash the original text and hand it back
|    BEFORE .facts.json is written.
|
|    rules.md Section B requires this: what was checked stays
|    in the record even when it leaves the page.
|
v
_normalize_misfiled_facts -- the DELIBERATE EXCEPTION.
A relocation moves text to the field it always belonged in, so
nothing is lost and the corrected record is the better one.
It is NOT reversed.
```

### 13.7 Guard integrity

`guardrails.md` names every guard by `module.symbol`. `test_guardrails_doc.py` fails the suite if any named symbol stops existing — which is what keeps the document honest. Several tests exist specifically to stop a guard rotting rather than to test a feature:

- malformed model replies are discarded;
- scrubbing never moves the Developer Score;
- a refusal leaves no half-written `.docx`;

- Section A never reaches an API request.

**Adding a guardrail:** put it in code, add its symbol to the right table in `guardrails.md`, and let `test_guardrails_doc.py` confirm it resolves. *A guard nobody has watched fail is only a comment: write the test that breaks it deliberately.*

---

## 14. Authentication and session architecture

Three CAPTCHA gates, **three different session models**. This is the single most operationally significant fact about the system.

| =====           |                           |                      |
|-----------------|---------------------------|----------------------|
| =====+          |                           |                      |
| PORTAL          | SESSION MODEL             | HUMAN COST PER RUN   |
| =====           |                           |                      |
| =====+          |                           |                      |
| MahaRERA        | Reusable guest token      | 1 solve, then 90 min |
| of              |                           |                      |
| session_auth.py | cached to disk for 90 min | free reuse across    |
| projects        |                           |                      |
|                 | (real TTL ~100 min)       |                      |
|                 | CAPTCHA gates the SPA     | 0 if --token or      |
| cache hit       |                           |                      |
|                 | detail page, NOT the API  |                      |
|                 |                           |                      |
| +-----+         |                           |                      |
| +-----+         |                           |                      |
| GST Portal      | Per-lookup                | 1 for the PAN search |
|                 |                           |                      |
| gst_portal.py   | One solve covers ALL      | + 1 per GSTIN        |
| discovered      |                           |                      |
|                 | financial years for one   |                      |
|                 |                           |                      |
|                 | GSTIN                     | 0 if --gstin/--pan   |
| omitted         |                           |                      |
| +-----+         |                           |                      |
| +-----+         |                           |                      |
| Maha Bhulekh    | *** NONE ***              | 1 per Property Card  |
| fetch           |                           |                      |
| mahabhumi.py    | The CAPTCHA image         |                      |
|                 |                           |                      |
|                 | regenerates on EVERY      | This is precisely    |
| why the         |                           |                      |
|                 | partial postback, so      | land-record path is  |
| manual,         |                           |                      |
|                 | token_cache's model does  | opt-in and NEVER     |
| auto-run        |                           |                      |

```

|                                     |
|                                     |
+=====+
=====+

```

HARD CONSTRAINT, stated in all three modules:  
 \*\*\* No module reads or solves a CAPTCHA image. \*\*\*  
 A human must look at the real browser window and type it in.

#### 14.1 The degradation ladder

```

--token X
  | miss
  v
90-minute disk cache
  | miss
  v
visible-browser CAPTCHA solve (300 s budget, 2.0 s poll)
  | timeout / window closed
  v
auth_source = "none" -> fetch ONLY projects + complaints
                        (the 2 categories needing no token at
all)
                        *** the run still completes ***

```

ON 401/403 FOR GATED CATEGORIES DESPITE HAVING A SESSION:  
 token\_cache.invalidate() -> re-solve -> refetch ONLY the failed  
 subset  
 \*\*\* exactly one retry attempt. No exponential backoff anywhere.  
 \*\*\*

#### 15. External integration architecture

| System          | Host / endpoint                                                   | Transport     | Auth                 | Failure policy                         |
|-----------------|-------------------------------------------------------------------|---------------|----------------------|----------------------------------------|
| MahaRERA search | maharera.maharashtra.gov.in                                       | Playwright    | none                 | ProjectNotFoundError → exit 1          |
| MahaRERA API    | maharera.it.maharashtra.gov.in                                    | requests POST | guest bearer, 7 of 9 | per-category isolation; one auth retry |
| MahaRERA DMS    | /api/maharera-dms-service/batch-job/downloadDocumentForPublicView | POST + bearer | yes                  | manifest row records the failure       |
| MahaRERA        | /orders-                                                          | Playwright +  | none                 | _safe_judgme                           |

| System          | Host / endpoint                         | Transport                      | Auth              | Failure policy                                      |
|-----------------|-----------------------------------------|--------------------------------|-------------------|-----------------------------------------------------|
| judgments       | judgements                              | retry                          |                   | nts_search – never fatal                            |
| ICRA            | public search + rating detail           | requests                       | none              | never fatal; one of two agencies                    |
| Infomercials    | public search + rating detail           | requests                       | none              | never fatal                                         |
| IBBI            | public insolvency search                | requests                       | none              | never fatal                                         |
| ZaubaCorp       | CIN redirect trick                      | requests                       | none              | _looks_paywa lled detection; chain continues        |
| Tofler          | type name in a real browser, verify CIN | Playwright                     | none              | chain continues                                     |
| InstaFinancials | ASP.NET web service, CIN search         | requests                       | none              | chain continues                                     |
| GST portal      | services.gst.gov.in                     | Playwright + Tesseract         | CAPTCHA/lookup    | graded note strings; never raises                   |
| Maha Bhulekh    | bhulekh.mahabhumi.gov.in                | Playwright + Tesseract mar+eng | CAPTCHA/call      | graded note strings; never raises                   |
| OSM Nominatim   | nominatim.openstreetmap.org/search      | requests, ≥1.1s apart          | none              | failed geocode is <b>dropped</b> , never “0 km”     |
| Google Maps     | consumer UI                             | Playwright                     | none              | <b>opt-in only</b> ; ToS caveat; always falls back  |
| Anthropic       | Claude API                              | anthropic SDK, tool_runner     | ANTHROPIC_API_KEY | every helper returns empty – deterministic fallback |
| xAI Grok        | https://api.x.ai/v1/grok-4.5            | OpenAI client                  | XAI_API_KEY       | optional second opinion on document grounding       |

## 15.1 Two integrations that carry explicit caveats

**Google Maps distance refinement.** COMPANY\_CHARTER\_USE\_MAPS\_SCRAPE must equal "1" exactly. It scrapes Google's *consumer UI* rather than the paid Distance Matrix/Routes API, so it may not comply with Google's Terms of Service and can break without warning. **Off by default.** Always falls back to the model's web\_search estimate. The template's own Methodology Note states that distances are estimated.

**Group companies.** Sourced from **ZaubaCorp only** — Tofler and InstaFinancials do not expose that data freely — and corroborated against InstaFinancials directorship counts rather than treated as authoritative.

---

## 16. LLM integration architecture and cost model

### 16.1 Single transport

EVERY Claude API call in the entire system funnels through:

```
deep_research._run_agentic_pass(user_prompt, system, label)
|
v
spent = usage_summary()["total"]["cost_usd"]    (read fresh,
never cached)
spent >= PIPELINE_COST_CAP_USD ($6.00) ?
|                                     \
| NO                                YES -> raise
CostCapExceeded
|                                     \
v                                     (never
retried -- a                             call
client.beta.messages.tool_runner(         flight
already in                               to
model = MODEL = "claude-sonnet-5",        call
is allowed                               START
max_tokens = 8000,                        cap is
finish; only a
system = <blocks>,
that would
tools = [{"type": "web_search_20260209",
after the
"name": "web_search"}],
refused)

messages = [{"role": "user", "content": user_prompt}],
cache_control = {"type": "ephemeral"})
|
| the runner is an ITERATOR -> one BetaMessage per turn
v
messages = list(runner)
|
+--> _record_usage(label, MODEL, messages)
```

```

because      |      sums input+output tokens across ALL turns,
--           |      each web_search round-trip is separately billed
--           |      counting only the last message would UNDERCOUNT.
cache_read_input_tokens |      cache_creation_input_tokens /
from         |      are read off usage too -- reported separately
--           |      input_tokens by the API, priced separately here
--           |      (_CACHE_WRITE_MULTIPLIER 1.25x,
_CACHE_READ_MULTIPLIER |      0.10x of the base input rate) rather than folded
in           |
--> _parse_json_response(messages[-1])
json.loads   |      strips ``` fences and a leading "json",
--           |
chars        |      failure -> RuntimeError with the first 500 raw

```

Structured output is enforced by PROMPT + PARSER, not by tool schema.

Only server tools are given -- raw tool-schema dicts auto-execute server-side only for recognised server tools, so no custom handler is registered, deliberately.

cache\_control is passed UNCONDITIONALLY on every call (2026-08-13). Per the SDK it marks the last cacheable block in the request -- system plus tools, which precede the varying user message -- so a system prompt reused across calls sharing one label (every batched verify/gap-retry/finding-research call does) is billed at the cache-read rate on its second and later uses. A one-off call still writes the cache once at a small premium; there is no case where this costs more.

## 16.2 Call-site inventory

| Caller                                                             | Label                        | Rules injected                                          |
|--------------------------------------------------------------------|------------------------------|---------------------------------------------------------|
| company_charter._run_<br>charter_pass                              | charter_pass                 | <b>Section B</b> (cacheable prefix)<br>+ _SYSTEM_PROMPT |
| company_charter._verify_one_field -<br>deep_research._verify_claim | material_claim_verify        | none                                                    |
| company_charter._verify_document_claim                             | document_grounding_verify    | none                                                    |
| company_charter._verify                                            | (not recorded — xAI, outside | none                                                    |

| Caller                                                                 | Label                                   | Rules injected                                                                                      |
|------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------|
| fy_document_claim_via_grok                                             | <i>the ledger</i> )                     |                                                                                                     |
| company_charter._llm_verify_citation_completeness                      | citation_completeness_judge             | <b>Section B + Section C</b> — the only call that exists specifically because a variant is External |
| company_charter._attempt_second_source                                 | second_source_verify                    | none                                                                                                |
| run_editorial_passes_pass1                                             | clean_check_judge                       | none                                                                                                |
| run_editorial_passes_pass2                                             | citation_match                          | none                                                                                                |
| run_editorial_passes_pass3                                             | flag_headline                           | none                                                                                                |
| run_finding_research                                                   | finding_research                        | none                                                                                                |
| run_claude_md_document_review                                          | claude_md_doc_review                    | <b>B</b> (internal) / <b>B + C</b> (external), passed as an argument                                |
| deep_research.run_deep_research                                        | research_generate                       | none                                                                                                |
| deep_research._verify_block – _verify_claims_batch                     | verify_claim_batch                      | none                                                                                                |
| deep_research._resolve_gaps – _retry_gaps_batch / _verify_claims_batch | gap_retry_batch, gap_retry_verify_batch | none                                                                                                |

**Section A never appears in this table.** That is the point.

### 16.3 Cost model

```

MODEL = "claude-sonnet-5"
_PRICING_PER_1M_TOKENS = {"claude-sonnet-5": {"input": 2.00, "output": 10.00}}
_CACHE_WRITE_MULTIPLIER = 1.25    # x base input rate, for cache_creation_input_tokens
_CACHE_READ_MULTIPLIER = 0.10     # x base input rate, for cache_read_input_tokens
PIPELINE_COST_CAP_USD = 6.0       # hard ceiling, checked in _run_agentic_pass

cost_usd = (
    input_tokens * price["input"]
    + cache_write_tokens * price["input"] * _CACHE_WRITE_MULTIPLIER
    + cache_read_tokens * price["input"] * _CACHE_READ_MULTIPLIER

```



```

    + output_tokens * price["output"]
) / 1_000_000

```

- An **unknown model** yields `{"input": 0.0, "output": 0.0}` so tokens still count but cost reads 0.0 rather than “a silently wrong number”.
- Per-label rollup: `usage_summary()` → `output/<reg>/usage_summary.json`; one rollup line appended to `output/usage_log.jsonl`. Each bucket now also carries `cache_write_tokens` and `cache_read_tokens`, summed separately from `input_tokens/output_tokens`.
- The Internal document’s **version log** carries elapsed time, total cost and call count for the whole end-to-end run — hence `pipeline_start_time` being threaded from `main.py`.
- **reset\_usage\_log()** is called at the start of every run because `app.py`’s Streamlit process is long-lived and would otherwise carry token counts across unrelated projects.
- **PIPELINE\_COST\_CAP\_USD (2026-08-13, \$6.00)** is a hard ceiling on total run spend — deep research and Charter generation combined, not just one stage — checked at the top of every `_run_agentic_pass` call by reading `usage_summary()["total"]` [`"cost_usd"`] fresh (never a stale snapshot). A call that would **start** at or over the cap raises `CostCapExceeded` and is **not retried** (unlike `BudgetExhausted`: a smaller search budget doesn’t make an already-overspent run cheaper); a call already in flight is allowed to finish, so the real ceiling is the cap plus at most one more call’s worst case. `CostCapExceeded` subclasses `RuntimeError`, so it degrades exactly like a missing `ANTHROPIC_API_KEY` — the stage it interrupts is [`never fatal`] in `main.py`, never the whole run.

⚠ **Pricing time-bomb.** The comment records this as “*Sonnet 5’s intro rate, active through 2026-08-31 — update to the standard \$3.00/\$15.00 after that date.*” As of this document’s date (13 Aug 2026) that leaves **18 days**. After expiry, every cost figure in `usage_summary.json`, `usage_log.jsonl`, the run summary and the Internal version log **under-reports by one third** until the constant is updated. This is unrelated to, and not fixed by, `PIPELINE_COST_CAP_USD` — the cap is denominated in the same (soon-to-be-stale) dollar figures, so it under-enforces by the same one third once the rate lapses.

## 16.4 Verification semantics

`deep_research._verify_claim` (still used directly by `company_charter._verify_one_field`, `label material_claim_verify`) and `deep_research._verify_claims_batch` (used by `_verify_block`, one call per chunk of up to `BATCH_VERIFY_CHUNK_SIZE` sources) both resolve every claim to one of four statuses, and the handling of the fourth is the interesting design decision:

| Status      | Meaning                           | Handling                              |
|-------------|-----------------------------------|---------------------------------------|
| confirmed   | Source re-checked and stands      | keep                                  |
| unsupported | Source does not support the claim | <b>drop</b> into gaps with the reason |
| stale       | Source is out of date             | <b>drop</b> into gaps with the        |

| Status             | Meaning                                                                                                                       | Handling                                                                                                          |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| verification_error | The <b>check itself</b> could not run (missing key, network, rate limit, bad JSON, or a chunk skipped because budget ran out) | reason<br><b>keep the source</b> , and append an honest gap: “(NOT independently re-verified this pass — reason)” |

An unrecognised status, or a claim id the model’s reply omits, is coerced to unsupported / treated as unverified. The verification\_error branch is what prevents an API outage — or a budget-exhausted chunk — from silently deleting good sourcing. Batching (2026-08-13) changed *how many calls* this costs, never *what a status means or how it’s handled* — the four-way semantics above are identical to the pre-batch, one-call-per-source implementation.

### 17. Scoring subsystem

Two scores coexist, with **deliberately opposite missing-data policies**.

#### 17.1 Developer Score — never renormalises

```
_compute_developer_score(facts, flags) -> {composite, grade, criteria}
```

| +                    | -----+                   |
|----------------------|--------------------------|
| BUCKET               | WEIGHT SUB-METRICS       |
| EACH                 |                          |
| +                    | -----+                   |
| Operational Strength | 50.0 Team Strength       |
| 12.5                 |                          |
|                      | Influence in Micromarket |
| 12.5                 |                          |
|                      | Past Experience - Area   |
| 12.5                 |                          |
|                      | Track Record             |
| 12.5                 |                          |
| +                    | -----+                   |
| Financial Strength   | 20.0 Financial Strength  |
| 20.0                 |                          |
| +                    | -----+                   |
| Governance Strength  | 30.0 RERA Compliance     |
| 7.5                  |                          |
|                      | GST Compliance           |
| 7.5                  |                          |
|                      | Cases (Past Defaults)    |



## conversation TODAY

```
tier: 0 -> AAA | <=20 -> AA | <=40 -> A | <=60 -> B | <=80 -> C |  
else D
```

The breakpoints are SHARED with `_classify_flags` via `_FLAG_THRESHOLDS`,  
so a flagged project cannot silently score AAA.

### TWO DISTINCT UNSCORED OUTCOMES:

```
no gst_compliance_check / not found -> a PENDING-INPUT gap  
on_time + late == 0                 -> "no period with both a  
                                     resolvable due date and a  
                                     recorded filing outcome"
```

Statutory due dates, computed offline in `gst_compliance.py`:

| Frequency                        | GSTR-1 | GSTR-3B                                                       |
|----------------------------------|--------|---------------------------------------------------------------|
| Monthly (span 28–31 days)        | 11th   | 20th                                                          |
| QRMP quarterly (span 89–92 days) | 13th   | <b>22nd</b> (Category X states) /<br><b>24th</b> (Category Y) |

Category X state codes: 22, 23, 24, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35, 36, 37 —  
Maharashtra is 27. “Not in Category X” is the correct test for Y; Y is deliberately not  
enumerated.

**Frequency is detected per *period*, not per GSTIN**, because a taxpayer can switch in or out of  
QRMP at the start of any quarter.

**Determinism warning.** `as_of` defaults to today. Callers generating scored, persisted  
Charter output must pass an explicit date, “otherwise the same `facts.json` re-  
rendered on a later date silently produces a different `delays_last_12_months`  
figure with no record of why.” `run_gst_compliance_check` does pass  
`as_of=datetime.now().date()` — which makes the *run* deterministic but not  
the *re-render*.

## 17.3 Documentation Confidence — always renormalises

Scores **this document’s sourcing**, not the promoter.

| Criterion                                | Weight |
|------------------------------------------|--------|
| <code>verification_rate</code>           | 0.25   |
| <code>source_tier_quality</code>         | 0.20   |
| <code>primary_tier_density</code>        | 0.15   |
| <code>cross_corroboration</code>         | 0.15   |
| <code>financial_figures_confirmed</code> | 0.10   |
| <code>completeness_rate</code>           | 0.05   |

| Criterion     | Weight |
|---------------|--------|
| recency_legal | 0.05   |
| recency_other | 0.05   |

N/A criteria are excluded and remaining weights **renormalised to 100 %** — the opposite policy to the Developer Score — except zero verification attempts, which has its own override. `_TIER_WEIGHTS` maps 12 source tiers to 20–100. `_REGENCY_WINDOW_MONTHS` = 18; `_REGENCY_WINDOW_MONTHS_LEGAL` = 3. Bands: High  $\geq 70$ , Moderate  $\geq 45$ , Limited below.

Both score tables **stay fully intact however many N/A rows they have** — they are scoring methodology, not fact tables, and are exempt from empty-row removal.

## 17.4 Flags

`_classify_flags` is fully deterministic — no LLM. It produces `{imminent[], structural[], monitor[]}` with items `{text, field[, gap_number]}`, and assigns every `gaps[i]` a **stable 1-based gap\_number** so a flag headline and its gap can be cross-referenced.

Checks: missing registration number · missing CTS/plot (imminent if `net_area` also missing, else structural) · **cts\_mismatch\_note — always imminent** · missing CIN/LLPIN · IBBI hit via `_classify_ibbi_hit` · complaint/appeal volume against `_FLAG_THRESHOLDS` · GST filing pattern · promoter portfolio total (always structural) · credit rating · mortgage\_lender\_history\_note (monitor).

```
_FLAG_THRESHOLDS = {
    complaint_monitor: 15,    complaint_imminent: 40,
    appeal_monitor: 5,       appeal_imminent: 15,
    credit_rating_min_units: 500,
    gst_late_pct_monitor: 15,    gst_late_pct_imminent: 40,
    gst_recent_delays_monitor: 1,  gst_recent_delays_imminent: 3,
}
```

**Flags summarise, Gaps explain.** `_flag_headline` renders a one-sentence headline ending in (Gap N); the full explanation lives once, under Gaps & Sources. A documented known limitation: on the 2026-08 Pranami data, 11 of 17 gaps were single sentences, so the headline was byte-identical to the gap — explicitly “*do not fix by truncating.*”

## 18. Document rendering architecture

### 18.1 The Internal / External fork

```
=====
=====
    _fill_template() -- ONE FUNCTION, TWO VARIANTS, ONE FACTS DICT
=====
=====
```

```

                                facts (real, mutated in place)
                                |
                                +-----+-----+
                                |               |
                                v               v
doc_variant = "internal"      doc_variant =
"external"                    |
                                |
                                |               |
_externalized_facts_copy()    *** DEEP COPY --
                                |               |
never                          touches the
                                |               |
real dict ***                  v
                                |               |
                                v               v
_preflight_rules("internal")  validates A + B
_preflight_rules("external")
+ C                             validates A + B
                                |               |
                                v               v
_normalize_misfiled_facts -> _scrub_clean_checks ->
_sanitize_process_gaps
(corrections FIRST, scrub SECOND -- the order matters)
                                |               |
                                v               v
shutil.copy2(TEMPLATE_PATH, out_path); python-docx opens the
COPY                            |               |
*** the template is filled IN PLACE, preserving every style,
colour                          |               |
                                v               v
and table already baked into it ***
                                |               |
                                v               v
monkey-patch doc.add_paragraph -> _add_paragraph_sanitized
BOTH: _strip_inline_cin_din + _expand_jargon_first_use
EXTERNAL ONLY: + _externalize_prose
                                |               |
                                v               v
"(label)" citations            "[N]" numbered
citations                      |
                                v               v
"(Code-Computed)" headings kept headings
stripped                       |
                                v               v
Document Library table t[8] kept t[8] DROPPED
entirely                       |
                                v               v
Weight columns kept           Weight columns
DROPPED                        |
                                v               v
ALL gaps printed              only gaps that
also                           earned an

```

Imminent or

flag,

"Gap N."

full bibliographic Sources list  
from

with [published:, accessed:]  
\_citation\_registry order

"Source Trust Registry Updates"  
version log (elapsed / \$ / calls)

8 internal-only paragraphs kept  
13, 17, 18,

DELETED

|  
v  
\*\_Internal.docx  
|  
\_verify\_external\_document\_quality  
|  
RuntimeError)  
|  
judge  
|

Structural

colour-coded,

Sources rebuilt

(suppressed)  
(suppressed)  
paragraphs 12,

36, 38, 58, 63

|  
v  
\*\_External.docx  
|

(11 checks ->

+ advisory citation

+-----+  
|

run\_claude\_md\_document\_review  
\*\*\* BLOCKS THE PDF \*\*\*

|  
v  
\_convert\_docx\_to\_pdf x2 (docx2pdf / Word COM)  
|  
v

\*\_Internal.pdf + \*\_External.pdf <-- THE

DELIVERABLE

|  
v  
\_restore\_clean\_checks + restore gaps  
|  
v

\*.facts.json <-- THE FULL RECORD

## 18.2 The ten locked top-level sections

**rules.md \$A: "Section order is locked. Ten top-level sections, same sequence, always. Reshape content inside the flow; never reorder or rename a section."**

| # | Heading         | Origin              |
|---|-----------------|---------------------|
| 1 | (title block) — | template paragraphs |

| #  | Heading                                                                                                          | Origin                                                                                                                                                                                                                                           |
|----|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | project/promoter line,<br>"Public Web-Sourced<br>Edition -- Adapted<br>for Maharashtra<br>(MahaRERA)", date line |                                                                                                                                                                                                                                                  |
| 2  | <b>Overview &amp; Flags</b>                                                                                      | _append_overview_section, relocated to the front                                                                                                                                                                                                 |
| 3  | <b>Executive Summary</b> + 4<br>KPI cards                                                                        | template +<br>_append_executive_summary_kpis                                                                                                                                                                                                     |
| 4  | <b>Counterparty</b>                                                                                              | renamed from "1. Legal<br>Identifiers...";<br>_append_counterparty_summary relocated in                                                                                                                                                          |
| 5  | <b>The Asset</b> (H2 child: <i>FSI</i><br>( <i>Floor Space Index</i> ))                                          | renamed from "2. Location<br>Map"; Land Identification<br>physically relocated in                                                                                                                                                                |
| 6  | <b>Compliance &amp; Legal Detail</b>                                                                             | renamed from "4. Rules";<br>gains <i>Complaint Order</i><br><i>Outcomes</i> and <i>Appeal-Level</i><br><i>Judgments</i> as H2                                                                                                                    |
| 7  | <b>Market &amp; Area<br/>Intelligence</b>                                                                        | renamed from "5. Area<br>Intelligence"                                                                                                                                                                                                           |
| 8  | <b>RERA Core Data</b>                                                                                            | renamed from "6. RERA<br>Scraping..."                                                                                                                                                                                                            |
| 9  | <b>Diligence Appendix</b>                                                                                        | new H1; H2 children:<br>Document Library Contents,<br>Credit Rating Check, IBBI<br>Insolvency Check, Company<br>Registration Profile, Group /<br>Affiliated Companies,<br>Developer Score,<br>Documentation Authenticity<br>& Confidence Summary |
| 10 | <b>Gaps &amp; Sources</b> (H2 child:<br><i>Sources</i> )                                                         | renamed from "Gaps &<br>Limitations..."                                                                                                                                                                                                          |

**Two conditional sections remain standalone Heading-1 appends at the document tail,**  
marked # unmoved in the code:

- Land Record Check -- Maha Bhulekh Property Card (Code-Assisted, Human-Verified)
- Review Authenticity Triage (Code-Computed Heuristics)



Both silently no-op unless the corresponding check ran, which is why they do not disturb the locked ten in the ordinary case. **This is a latent inconsistency with the locked-order rule and should be recorded as such** (see §25).

### 18.3 Tables

| Index | Table               | Notes                                                                                |
|-------|---------------------|--------------------------------------------------------------------------------------|
| t[0]  | Land Identification | rows 1–7 + a grown land_assembly row                                                 |
| t[1]  | Corporate Identity  | rows 1–9; one of only three tables where a CIN/DIN may appear inline                 |
| t[2]  | Neighbourhood       | east / west / north / south                                                          |
| t[3]  | Distances           | variable                                                                             |
| t[4]  | FSI Metrics         | rows 1–5 + the merged "Mortgage / charge on the land" row via _merged_mortgage_value |
| t[5]  | Comparables         | variable                                                                             |
| t[6]  | RERA Core Data      | rows 1–13                                                                            |
| t[7]  | Blocks              | variable                                                                             |
| t[8]  | Document Library    | <b>Internal only</b>                                                                 |

### 18.4 Typographic rules enforced in code

| Rule                                                                                | Implementation                                                                                                                                                                       |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Body text justified                                                                 | _justify_body_paragraphs                                                                                                                                                             |
| Table cells centred <b>for short values only</b>                                    | _center_all_table_cells with a length threshold — the original unconditional version produced unreadable Landowner/Investor, Board Resolution and Mortgage rows (amended 2026-08-10) |
| Table rows must not split across a page break; a continued table repeats its header | _apply_table_pagination sets cantSplit on each row and tblHeader on the header row                                                                                                   |
| No URLs in narrow table columns                                                     | cells carry a short label or an [N] marker; the full URL lives once in Sources                                                                                                       |
| Bullet hanging indent                                                               | _fix_bullet_hanging_indent + _apply_bullet_hanging_indent; gate check 11 verifies abstractNum id 2                                                                                   |
| No manual page-break characters                                                     | headings use page_break_before=True — a manual break produced a real blank-page bug (enforced in charter_report's gate)                                                              |

## 18.5 Two pieces of hidden state

`_fill_template` sets a module-global `_ACTIVE_EXTERNAL_FACTS` and monkey-patches `doc.add_paragraph`. Both are reset per call, but together they make **`_fill_template` non-reentrant and not thread-safe**. It must not be parallelised across variants.

## 18.6 The cheap verification loop

`_fill_template` runs straight off a saved `facts.json` and makes **no API calls**, so most rendering behaviour can be verified for free:

```
import json, company_charter as cc
facts =
json.load(open('output/company_charters/Company_Charter_<Name>_<REG>.facts.json',
               encoding='utf-8'))
cc._fill_template('<REG>', facts,
                  'output/company_charters/_verify/int.docx',
                  doc_variant='internal')
cc._fill_template('<REG>', facts,
                  'output/company_charters/_verify/ext.docx',
                  doc_variant='external')
```

Any script hitting `_fill_template` directly **must call `_convert_docx_to_pdf` itself afterwards**. `run_company_charter` is the only path that does it for you.

---

## 19. Secondary and standalone builders

```
=====
DOCUMENT FAMILIES, BUILDERS AND RENDERERS
=====
```

```
FAMILY 1 -- RERA PROJECT SUMMARY
+-----+
| report.py :: build_pdf()          ReportLab          |
| LIVE. Stage 10 of main.py; also finalize_report.py, app.py. |
| -> output/<REG>/<REG>_summary.pdf                      |
+-----+

FAMILY 2 -- COMPANY CHARTER
+-----+
| (a) company_charter.py :: _fill_template()  python-docx |
| *** THE LIVE AUTOMATIC PATH ***                  |
| via run_company_charter, called from main.py        |
| -> Internal + External .docx -> .pdf                |
| Gate: _verify_external_document_quality (11 checks)  |
| (b) charter_report.py :: build_charter_report()  python-docx |
+-----+
```

MANUAL / STANDALONE ONLY. Never called from main.py.  
 "Counterparty + Collateral" restructured document.  
 Own already-implemented findings-not-absence policy.  
 Gate: verify\_charter\_report\_quality (6 checks, incl. the  
       bare-domain check the other gate does NOT have)  
 Does NOT convert to PDF -- conversion is a separate step.

(c) executive\_briefing.py :: build\_executive\_briefing()  
 python-docx + matplotlib (Agg) donut charts.  
 8-page narrative COMPANION, not a replacement.  
 Narrates only what the Charter already established;  
 performs no new interpretation of raw data.  
 No \_\_main\_\_ and no in-repo caller -- a host must import it.

(d) charter\_document.py  
 \*\*\* NOT A BUILDER. A LIVE SHARED LIBRARY. \*\*\*  
 build\_charter\_document() and the 24 definitions only it  
 reached (the \_Builder class + all 15 \_section\_N\_\* render-  
 ers) were DELETED 2026-08-10 after an audit confirmed it  
 had never been called from a non-test file since 5b8c6ee.  
 ~710 lines remain. charter\_report.py imports 16 symbols;  
 executive\_briefing.py uses \_green\_flags.  
 \*\*\* DO NOT DELETE. DO NOT TREAT AS DEAD CODE. \*\*\*  
 (CHARTER\_RESHAPE\_SPEC.md calls it dead code. That file is  
 stale and must not be followed.)

## PDF CONVERSION

company\_charter.\_convert\_docx\_to\_pdf -- docx2pdf / Word COM  
 The ONLY converter in the system. Windows-only.  
 run\_company\_charter returns the INTERNAL PDF path as out\_path,  
 falling back to the .docx only if conversion failed.  
 \*\*\* THE PDF IS THE DELIVERABLE, NOT THE .DOCX \*\*\*

### 19.1 charter\_report.py — the second builder

Section structure ("Counterparty + Collateral"):

Cover page ---- COMPANY CHARTER (28 pt navy) / "Company, Promoters  
 and

Collateral / Collateral Due-Diligence Note" / Company /

Location / Date of review / classification line  
 ("Internal -- Integrow Asset Management" vs  
 "Strictly Private and Confidential")  
 Composite Developer Score: N/100 (grade G)

1. About the Company ..... prose only, no tables
2. The Company ..... Identity and Registration

|                                                                                                   |                                                                                                                                           |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Documentation                                                                                     | Verification Summary<br>(Overall Rating,                                                                                                  |
| Verify)                                                                                           | Confidence, Key Items to                                                                                                                  |
| Screening                                                                                         | Portfolio and Track Record<br>Litigation and Regulatory                                                                                   |
| 3. The Promoters .....<br>director                                                                | >>> Needs Attention, Company<br>one subsection per current<br><br>+ one material past director<br>>>> Needs Attention,                    |
| Promoters<br>4. The Collateral: <project> .....<br>Record                                         | Asset Identity and Land<br><br>Approvals, FSI and Title<br>RERA Compliance and Escrow<br>Project Professionals<br>Location, Connectivity, |
| Market Read                                                                                       | Document and Diligence Trail<br>Litigation and Regulatory                                                                                 |
| Screening                                                                                         | >>> Needs Attention,                                                                                                                      |
| Collateral<br>5. Closing Read .....<br>verdict                                                    | signals/finding table +<br><br>Recommended Verification                                                                                   |
| Steps<br>6. Scoring Detail (Appendix) .....<br>Annexure: Related Entity Mapping ....<br>promoters | by sub-metric, by criterion<br>one combined table for all                                                                                 |
| References .....<br>inline)                                                                       | EXTERNAL ONLY (Internal uses<br>self-descriptive "(per X)"                                                                                |

\*\*\* There is no "What Checks Out" heading anywhere in this document. \*\*\*

Every attention item is consolidated into that section's own "Needs Attention, <label>" subsection.

Its gate, `verify_charter_report_quality(docx_path)`, walks every body paragraph **and every table cell recursively** on the re-opened saved file:

1. Dash artefacts — " - - ", en dash, em dash surviving `_clean_text`
2. Redundant inline attribution — a paragraph matching both a citation marker and a "(per X)" / "confirmed identically across..." pattern
3. **Bare-domain mentions** never converted into a citation — *the check the other gate does not have*

4. No manual page-break character (w:br w:type="page") — headings must use page\_break\_before=True
5. Footer page-number fields — both PAGE and Numpages w:instrText must be present
6. Citation integrity — reports missing\_refs and orphan\_refs separately

## 19.2 The orchestrator seam

run\_charter\_pipeline.py

```
+-----+
|   prepare()  -> charter_research_prep.build_roster(facts)
|
|               build_director_prompt() / build_group_prompt()
|
|               RETURNS PROMPTS
|
|
|
|               >>> the HOST (an LLM orchestrator) runs the research
|
|               agents in parallel -- spawning a research agent is a
|
|               tool call only an orchestrator can make <<<
|
|
|
|   finish()   -> parse_agent_json() -> assemble_research()
|
|               -> charter_report.build_charter_report()
|
|               -> (PDF conversion is a separate post-finish step)
|
+-----+
```

build\_report.py is the hand-transcribed equivalent of that middle step for one engagement. It is explicitly "not a reusable pipeline component -- delete or rewrite per engagement."

DEFAULT\_PAST\_DIRECTOR\_LINK\_THRESHOLD = 5: a past director needs at least 5 of their own related-entity links to earn a subsection; the rest become additional\_next\_steps. is\_partnership swaps director↔partner throughout, after confirmed live bugs for an LLP promoter and a bare partnership firm with no DIN — the no-DIN branch changes the whole search strategy.

## 19.3 Pre-RERA intake paths

A CIN or a CTS number often reaches the pipeline BEFORE a RERA registration number exists. Two standalone intakes handle that:

promoter\_intake.py <CIN> [company\_name]

```

-> MCA profile + IBBI + group companies + credit rating
-> output/_pending/<CIN>/promoter_profile.json
    *** written with the EXACT facts.json keys ***

cts_intake.py <district> <office> <village> <cts> <mobile>
-> output/_pending/<district_village_cts>/land_record.json
    key: cts_land_record_check -- again the exact facts.json
key
    (a bare CTS number is not globally unique;
district+village+
    cts_number together form the key)

|
| later, when a RERA number is known:
v
attach_rera.py <case_id> <reg_no>
-> COPIES (never moves) into output/<reg_no>/ as
*_carryover.json
-> a later `python main.py <reg_no>`:
    reuses the promoter profile INSTEAD of re-running live
CIN
    checks (the 5-way fan-out skips four of its five
branches)
    reuses the land record INSTEAD of opening a new CAPTCHA
    browser, and cross-checks the CTS against RERA's own
    survey_cts_plot_numbers
    *** a genuine mismatch -> facts["cts_mismatch_note"]
    -> _classify_flags promotes it to IMMINENT ***

"an explicit action, never auto-matched" -- the pipeline's standing
policy that a human confirms every identifier link.

```

---

## 20. Concurrency and performance

### 20.1 Concurrency map

| COMPONENT                                                        | MODEL                    |   |
|------------------------------------------------------------------|--------------------------|---|
| WORKERS                                                          |                          |   |
| -----                                                            |                          |   |
| -                                                                |                          |   |
| fetch_all_categories                                             | ThreadPoolExecutor       | 8 |
| (projects fetched serially first -- the one ordering dependency) |                          |   |
| download_documents                                               | ThreadPoolExecutor       | 8 |
| download_complaint_orders                                        | SERIAL                   | 1 |
| company_charter phase-2 checks                                   | ThreadPoolExecutor       | 5 |
| promoter_portfolio                                               | SERIAL                   | 1 |
| deep_research                                                    | SERIAL                   | 1 |
| gst_intake / gst_portal                                          | SERIAL (visible browser) | 1 |

|                           |                          |   |
|---------------------------|--------------------------|---|
| mahabhumi                 | SERIAL (visible browser) | 1 |
| _fill_template            | SERIAL, NON-REENTRANT    | 1 |
| discover.verify_endpoints | SERIAL (deliberate)      | 1 |

**Thread-safety notes.** - Every category and document worker creates **its own requests.Session()** — the library does not guarantee Session thread-safety, and the connection-reuse benefit of sharing one is marginal next to the risk. - One `threading.Lock` guards the only genuinely shared state in the download pool: `seen_filenames` and `seen_urls`. The check-then-add is the sole critical section. - `fetch_all_categories` consumes futures in submission order on the main thread, so the results dict is only ever mutated single-threaded. - `promoter_portfolio` uses a **module-global** `_last_geocode_at` to honour Nominatim's ~1 req/s policy — not thread-safe, by design, because the module is single-threaded. - `deep_research._USAGE_LOG` and its lazily-created singleton client are process-global.

## 20.2 Measured bottlenecks

`fetch_all_categories`' own docstring names its concurrency as *"the other half of a real, measured bottleneck (alongside the Phase 2 checks and MahaRERA judgments search) in a full pipeline run."*

The three measured hot spots, therefore:

1. Category fetch (parallelised, 8 workers)
2. Charter phase-2 external checks (parallelised, 5 workers)
3. MahaRERA judgments search (Playwright, with its own retry)

**LLM cost profile.** The docstring records that *"a handful of small calls (one per cited source needing re-verification) dominates far more than the one big Charter-assembly pass."*

Verification, not generation, is the cost centre.

**Human wall-clock** is the dominant end-to-end factor on any run using GST or land records: one CAPTCHA solve for MahaRERA, one for the GST PAN search, one per GSTIN, and one per Property Card fetch.

## 20.3 Timeout and bound constants (single reference)

| Constant                       | Value | Where                 |
|--------------------------------|-------|-----------------------|
| REQUEST_TIMEOUT                | 30 s  | all HTTP              |
| SEARCH_TIMEOUT_MS              | 30000 | Playwright navigation |
| CAPTCHA_TIMEOUT_SECONDS        | 300   | all three portals     |
| CAPTCHA_POLL_INTERVAL_SECONDS  | 2.0   | all three portals     |
| token_cache.MAX_AGE_MINUTES    | 90    | vs ~100 min real TTL  |
| _MAX_DOCUMENT_DOWNLOAD_WORKERS | 8     | api_client            |

| Constant                                            | Value                         | Where                   |
|-----------------------------------------------------|-------------------------------|-------------------------|
| PROMOTER_PROJECT_LIMIT                              | 25                            | config                  |
| _GEOCODE_MIN_INTERVAL_S                             | 1.1                           | Nominatim politeness    |
| _FIVE_KM_RADIUS                                     | 5.0 km                        | portfolio filter        |
| Earth radius                                        | 6371.0 km                     | haversine               |
| MAX_GAP_RETRY_ATTEMPTS                              | 2                             | deep_research           |
| MAX_FINDING_RESEARCH_CALLS                          | 8                             | deep_research           |
| MAX_RESEARCH_VERIFICATION_CALLS                     | 15                            | deep_research           |
| BATCH_VERIFY_CHUNK_SIZE                             | 10                            | deep_research           |
| BATCH_VERIFY_MAX_SEARCHES                           | 10                            | deep_research           |
| BATCH_GAP_RETRY_MAX_SEARCHES                        | 10                            | deep_research           |
| PIPELINE_COST_CAP_USD                               | \$6.00                        | deep_research           |
| _CACHE_WRITE_MULTIPLIER /<br>_CACHE_READ_MULTIPLIER | 1.25x / 0.10x base input rate | deep_research           |
| RESEARCH_REUSE_WINDOW_HOURS                         | 24                            | deep_research           |
| max_tokens                                          | 8000                          | every agentic pass      |
| review input cap                                    | 60000 chars                   | _rendered_document_text |
| _MIN_FINDING_LENGTH                                 | 80                            | company_charter         |
| _SOURCE_PROMOTION_HIT_THRESHOLD                     | 5 distinct projects           | trust_registry          |
| DEFAULT_PAST_DIRECTOR_LINK_THRESHOLD                | 5                             | charter_research_prep   |
| _LONG_VALUE_THRESHOLD                               | 300 chars                     | report.py               |
| _MAX_TABLE_COLS                                     | 6                             | report.py               |

There is **no exponential backoff anywhere in the system**. The only retry is `main.py`'s single auth-refresh, plus bounded per-attempt retries in `mahabumi` (3) and the judgments search.



## 21. Failure, degradation and recovery model

### 21.1 The degradation matrix

| =====                         |                         |           |
|-------------------------------|-------------------------|-----------|
| =====+                        |                         |           |
| FAILURE<br>OUTCOME            | COST                    | RUN       |
| =====                         |                         |           |
| =====+                        |                         |           |
| CAPTCHA not solved / timeout  | 7 of 9 categories       | COMPLETES |
| Token expired mid-run (401)   | one auth retry, then    | COMPLETES |
|                               | those categories only   |           |
| One category endpoint fails   | that category           | COMPLETES |
| One document download fails   | a manifest row          | COMPLETES |
| Complaint-order download dies | the whole sub-feature   | COMPLETES |
| Geocode fails for a project   | that entry dropped from | COMPLETES |
|                               | the 5 km sum (never     |           |
|                               | counted as "0 km away") |           |
| Promoter portfolio build dies | one Charter input       | COMPLETES |
| GST portal outage / no solve  | ONE unscored sub-metric | COMPLETES |
| ANTHROPIC_API_KEY missing     | deep research + Charter | COMPLETES |
|                               | (retry standalone)      |           |
| Deep research pass fails      | market/promoter context | COMPLETES |
| Editorial pass fails          | deterministic fallback  | COMPLETES |
| Per-finding research fails    | ONLY that finding, and  | COMPLETES |
|                               | its ORIGINAL TEXT KEEPS |           |
| Tesseract not installed       | "[OCR unavailable]"     | COMPLETES |
|                               | marker per document     |           |
| Google Maps scrape fails      | falls back to estimate  | COMPLETES |
| PIPELINE_COST_CAP_USD reached | whichever deep research | COMPLETES |

```

|
|                                     | / Charter call was next |
|
+-----+
| rules.md missing / malformed | *** BLOCKS *** |
RuntimeError |
| B or C carries an em dash | *** BLOCKS *** |
RuntimeError |
| External gate violation | *** BLOCKS THE SAVE *** |
RuntimeError |
| Verified rules violation | *** BLOCKS THE PDF *** |
CharterCompl... |
| Review could not run at all | *** BLOCKS THE PDF *** |
CharterCompl... |
| Template file missing | *** BLOCKS *** |
FileNotFound |
+-----+
| Project not found in search | run cannot start | exit(1)
|
| Multiple matches, no TTY | run cannot start | exit(2)
|
+=====+
=====+

```

## 21.2 Recovery paths

| Situation                                         | Recovery                                                                                                                                                   |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deep research failed                              | python deep_research.py <REG_NO><br>— reads raw/, rebuilds the PDF                                                                                         |
| Charter failed                                    | python company_charter.py<br><REG_NO>                                                                                                                      |
| Only the PDF needs rebuilding                     | python finalize_report.py<br><REG_NO> — <b>zero network calls, no Playwright, no CAPTCHA</b>                                                               |
| Search index cannot find a live project           | --project-id N — the direct API can still return full data for a project the search box cannot find; a site-side index gap, not a sign the project is gone |
| Endpoint drift suspected                          | python main.py <REG_NO> --verify<br>— probes every endpoint, prints confirmed/observed next to the empirical outcome                                       |
| A run needs to be diffed against the previous one | it already is:<br>output/_history/<reg>/<ts>/                                                                                                              |
| Rendering change needs checking                   | the cheap no-API_fill_template loop (\$18.6)                                                                                                               |

| Situation               | Recovery                                                                      |
|-------------------------|-------------------------------------------------------------------------------|
| Cached session is stale | <code>python token_cache.py get \   set &lt;token&gt; \   minutes-left</code> |

### 21.3 Known verify-mode blind spot

`discover.verify_endpoints` does not pass a body override, so `past_experiences` legitimately reports `FAILED: HTTP 400` even though the real pipeline handles it correctly via `_past_experiences_body`. Operators must not read that as a regression.

## 22. Security, privacy and compliance

### 22.1 Secrets

| Secret               | Storage                                                                                                  | Risk                | Mitigation                                            |
|----------------------|----------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------|
| MahaRERA guest token | <code>.guest_token_cache.json</code> , <b>plaintext</b> , <b>default permissions</b> , beside the source | Low                 | Short-lived (~100 min), guest-scope, public data only |
| ANTHROPIC_API_KEY    | environment variable                                                                                     | Medium              | Never written to disk by this system; never logged    |
| XAI_API_KEY          | environment variable                                                                                     | Medium              | Optional; absent — that fallback is simply skipped    |
| Promoter PAN / GSTIN | passed on the CLI; echoed into <code>output/&lt;reg&gt;/gst_portal_raw/</code> filenames and JSON        | <b>Medium — PII</b> | See §22.3                                             |

### 22.2 Ethical and legal boundaries — explicitly recorded in code

1. **No CAPTCHA is ever read or solved by the system.** Stated in `session_auth.py`, `gst_portal.py` and `mahabhumi.py`. A human solves every one.
2. **Google Maps scraping is opt-in and carries a ToS caveat.** `COMPANY_CHARTER_USE_MAPS_SCRAPES` defaults off; the code comment states plainly that it scrapes the consumer UI rather than the paid API and “may not comply with Google’s Terms of Service.”
3. **Nominatim is rate-limited to ~1 req/s** with a descriptive User-Agent identifying the tool as a low-volume personal research tool.
4. **DMS requests set Origin and Referer** mimicking the real detail page. This is anti-bot accommodation, not authentication bypass — the endpoints are public and require the guest token the site itself mints.

5. **Group-company data is taken only from the source that publishes it freely** (ZaubaCorp); paywalled mirrors are detected (`_looks_paywalled`) rather than circumvented.

### 22.3 Data protection observations

- **output/ is gitignored**, which is what keeps promoter PII, downloaded documents and GST filing data out of version control. This is load-bearing.
- **source\_trust\_registry.json IS committed**. It contains open-web domains and hit counts, not project data — acceptable, and desirable, because it makes trust promotions reviewable in version control.
- **.facts.json is the unscrubbed record** — it retains process failures, file paths and clean-check text that the External document deliberately strips. It must be treated as an **Internal-classification artefact** and never forwarded to a counterparty.
- **gst\_portal\_raw/ contains full-page screenshots** of GST portal responses keyed by PAN and GSTIN. These are PII-bearing and are retained indefinitely.
- **output/\_history/ grows without bound**. No retention policy exists.

### 22.4 Document classification

| Artefact           | Classification                       | Distribution            |
|--------------------|--------------------------------------|-------------------------|
| *_External.pdf     | Client-shareable                     | Counterparty / investor |
| *_Internal.pdf     | Internal — Integrow Asset Management | Internal only           |
| *.facts.json       | <b>Internal, unscrubbed</b>          | Never leaves the system |
| <REG>_summary.pdf  | Internal working document            | Internal                |
| output/<reg>/raw/* | Internal                             | Internal                |
| gst_portal_raw/*   | <b>Internal, PII-bearing</b>         | Internal                |

The External document's own classification line reads **"Strictly Private and Confidential"**; the Internal one reads **"Internal – Integrow Asset Management"**.

## 23. Testing architecture

### 23.1 Shape

28 `test_*.py` files at repository root. No `tests/` package, no `conftest.py`, no CI configuration, no `pyproject.toml`. Run with `python -m pytest -q`.

### 23.2 Three kinds of test

#### (1) FEATURE TESTS

Ordinary behaviour: scoring bands, parsing, portfolio maths.  
e.g. `test_developer_score.py`, `test_gst_compliance.py`,  
`test_promoter_portfolio.py`

#### (2) POLICY TESTS

Assert an EDITORIAL RULE holds in rendered output.  
e.g. `test_clean_check_scrubber.py` -- the Pranami litigation  
bullet must NOT survive, while the Lis Pendens finding MUST, and  
the Developer Score RERA-Compliance note must be UNTOUCHED.  
`test_external_citations.py`, `test_flag_gap_pointers.py`

(3) GUARD-INTEGRITY TESTS <-- the unusual category  
Exist specifically to stop a guard rotting, not to test a  
feature.

- . `test_guardrails_doc.py` -- fails if any symbol NAMED IN  
guardrails.md stops existing
- . malformed model replies are discarded
- . scrubbing never moves the Developer Score
- . a refusal leaves no half-written .docx
- . Section A never reaches an API request -- and this test now  
asserts the marker is PRESENT before asserting it is  
ABSENT,

after it was found PASSING VACUOUSLY

That last detail is the most instructive thing in the test suite: a guard test that passes for the wrong reason is worse than no test, and the fix was to assert the precondition first.

### 23.3 The cheap verification loop

Because `_fill_template` makes no API calls when run off a saved `facts.json`, the majority of rendering and editorial behaviour is testable **for free**. Only per-finding research needs live API calls.

### 23.4 Known fixture fragility

`CHARTER_RESHAPE_SPEC.md` Task 0 documents a fixture-naming drift that once broke 16 tests, 13 of them on Windows too, with six further failures cascading because one test builds a scratch file later tests consume. The lesson recorded there stands: `docx2pdf` is Windows-only and those tests need `pytest.importorskip("docx2pdf")` to keep the suite runnable on Linux/CI.

---

## 24. Deployment and operations

### 24.1 Runtime requirements

| Requirement | Detail                                                                                                                    |
|-------------|---------------------------------------------------------------------------------------------------------------------------|
| Python      | 3.11+ (uses <code>str \</code>   <code>None</code> syntax throughout)                                                     |
| OS          | <b>Windows required for the PDF deliverable</b> ( <code>docx2pdf</code> – Word COM).<br>Everything else is cross-platform |

| Requirement | Detail                                                                                                             |
|-------------|--------------------------------------------------------------------------------------------------------------------|
| Browser     | Chromium via playwright install chromium                                                                           |
| OCR         | Tesseract with eng + mar; tesseract/ is bundled in-repo. TESSERACT_CMD or the two standard Windows installer paths |
| Word        | Microsoft Word installed, for COM automation                                                                       |
| Network     | Outbound HTTPS to the systems in \$15                                                                              |
| Human       | Present at the keyboard for CAPTCHA solves                                                                         |

requirements.txt:

```

playwright>=1.44      requests>=2.31      reportlab>=4.0
streamlit>=1.32       anthropic>=0.40     python-docx>=1.1
pymupdf>=1.24         pytesseract>=0.3    Pillow>=10.0
beautifulsoup4>=4.12 docx2pdf>=0.1.8

```

matplotlib is used by executive\_briefing.py but is **not listed** — an undeclared dependency (see §25).

## 24.2 Environment variables

| Variable                         | Consumer                                        | Effect if unset                                                                                                                                                                |
|----------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ANTHROPIC_API_KEY                | deep_research (implicitly, via Anthropic())     | Deep research and Charter generation fail — never fatally                                                                                                                      |
| XAI_API_KEY                      | _verify_document_claim_via_grok                 | The Grok second-opinion path is skipped                                                                                                                                        |
| TESSERACT_CMD                    | company_charter, gst_portal, mahabhum           | Falls back to PATH, then two Windows installer paths, then "[OCR unavailable]"                                                                                                 |
| TESSDATA_PREFIX                  | set by the code to the project-local tesseract/ | mahabhum <b>overwrites</b> ; gst_portal uses setdefault — because TESSDATA_PREFIX <i>replaces</i> the search path, eng.traineddata must be copied in alongside mar.traineddata |
| COMPANY_CHARTER_USE_MAPS_SCRAPES | _refine_distances_with_maps                     | Must equal "1" exactly; otherwise the whole Maps upgrade is skipped                                                                                                            |
| CHARTER_ALLOW_UNCHECKED          | run_company_charter                             | 1/true/yes downgrades the blocking review to                                                                                                                                   |



24.4 Operational checklist — before calling a run done

Straight from CLAUDE .md:

- ☒ Correct builder used: `company_charter.py::_fill_template` via `run_company_charter.charter_report.py` is a different document for a different request; `charter_document.py` builds nothing at all any more.
  - ☒ A **real** monitor-flag resolution pass — flags re-checked against current facts, not carried over stale.
  - ☒ The review stage ran, or its failure was **reported rather than passed over**.
  - ☒ Output only in `output/company_charters/`.
  - ☒ Any script hitting `_fill_template` directly called `_convert_docx_to_pdf` itself.
- 

25. Known limitations, technical debt and risks

25.1 Deliberate, documented, do-not-fix

| Item                                                                                                                    | Why it stays                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Group/Affiliated Companies count contradiction</b> — 65 linked entities vs “of 299” in the Director Relationship Map | A real disagreement in the source data. Silently choosing one side would present a guess as a fact                                                                                                        |
| <b>Internal document library’s 61 rows</b> — 10 labelled only Other – Legal, 8 filenames appearing twice                | Same reason                                                                                                                                                                                               |
| <b>Developer Score never renormalises</b>                                                                               | A promoter with less publicly-verifiable data <i>should</i> structurally score lower                                                                                                                      |
| <b>Flag headlines byte-identical to their gap</b> on single-sentence gaps (11 of 17 on Pranami)                         | Explicitly “ <i>do not fix by truncating</i> ”                                                                                                                                                            |
| <b>charter_document.py looks like dead code</b>                                                                         | It is a live shared library. 16 symbols. Do not delete                                                                                                                                                    |
| <b>_verify_external_document_quality</b> treats “Document Library” as a violation                                       | Vestigial from when the section was Internal-only. Must be <b>narrowed</b> to the leftover template placeholder text before an External document library ships — otherwise every External save hard-fails |

25.2 Technical debt

| #  | Item                                                                                  | Impact                                                                                      | Suggested action                                                                                                                  |
|----|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| D1 | <b>company_charter.py is 10,182 lines / 591 KB</b> with six distinct responsibilities | Severe cognitive load; merge-conflict magnet; the file’s own analysis needs tooling to read | Extract, in order of independence: registry lookups – <code>registries.py</code> ; editorial passes – <code>editorial.py</code> ; |



| #  | Item                                                                                                                                            | Impact                                                             | Suggested action                                                                                                                               |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                                                                                                                                 |                                                                    | scoring –<br>scoring.py;<br>rendering primitives<br>–<br>docx_helpers.py.<br>Keep<br>_fill_template<br>and<br>run_company_char<br>ter together |
| D2 | <b>No packaging manifest and no CI</b>                                                                                                          | The suite is run by hand; drift is invisible until someone runs it | Add pyproject.toml + a GitHub Actions job running pytest -q with docx2pdf skipped                                                              |
| D3 | <b>matplotlib is an undeclared dependency</b>                                                                                                   | executive_briefing.py fails on a clean install                     | Add to requirements.txt                                                                                                                        |
| D4 | <b>Sonnet 5 intro pricing expires 2026-08-31</b>                                                                                                | Every cost figure under-reports by a third afterwards              | Update _PRICING_PER_1M_TOKENS to \$3.00/\$15.00; better, add a dated pricing table                                                             |
| D5 | <b>docx2pdf makes the deliverable Windows-only</b>                                                                                              | No Linux/container deployment path                                 | Evaluate LibreOffice --headless --convert-to pdf as a fallback converter                                                                       |
| D6 | <b>Two conditional sections sit outside the “locked ten”</b>                                                                                    | Latent inconsistency with a Section A rule                         | Either fold them into Diligence Appendix as H2, or amend the rule to say “ten <i>unconditional</i> sections plus two conditional appends”      |
| D7 | <b>fetch_gstin_filing_table’s docstring is stale</b> — it claims it does not parse into the record shape; parse_filing_table does exactly that, | Misleads a reader                                                  | Update the docstring                                                                                                                           |

| #   | Item                                                                                                            | Impact                                                                                        | Suggested action                                                                     |
|-----|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|     | confirmed against 76 periods                                                                                    |                                                                                               |                                                                                      |
| D8  | <b>gst_compliance.summarize_filing_pattern as_of defaults to today</b>                                          | Re-rendering the same facts.json later silently produces a different delays_last_12_months    | Persist the as_of used into facts["gst_compliance_check"] and re-use it on re-render |
| D9  | <b>output/_history/ and gst_portal_raw/ grow without bound</b>                                                  | Disk; PII retention                                                                           | Define and implement a retention policy                                              |
| D10 | <b>token_cache has no atomic write or file lock</b>                                                             | A concurrent save/load can observe a truncated file (degrades to “no token”, so low severity) | Write-to-temp-then-rename                                                            |
| D11 | <b>run_archive is not safe for two concurrent runs of the same reg-no</b>                                       | Timestamp collision check is not atomic                                                       | Advisory lock file per reg-no                                                        |
| D12 | <b>_fill_template is non-reentrant</b><br>(module-global _ACTIVE_EXTERNAL_FACTS + monkey-patched add_paragraph) | Cannot be parallelised across variants                                                        | Document it (done here); consider a context object instead of a global               |
| D13 | <b>executive_briefing.py has no caller and no __main__</b>                                                      | Unreachable except by import                                                                  | Add a CLI, or record it as library-only in CLAUDE.md                                 |
| D14 | <b>build_report.py hardcodes one engagement's paths</b>                                                         | Will fail for any other project                                                               | It says so itself — “delete or rewrite per engagement”                               |
| D15 | <b>Grok verification calls are not recorded in the usage ledger</b>                                             | Cost blind spot                                                                               | Route through _record_usage with its own label and pricing entry                     |

### 25.3 Operational risks

| Risk                                     | Likelihood | Impact                               | Mitigation in place                                                 | Gap                                      |
|------------------------------------------|------------|--------------------------------------|---------------------------------------------------------------------|------------------------------------------|
| MahaRERA changes an endpoint or selector | High       | Run fails or silently under-collects | confirmed/observed tiering; - - verify mode; envelope normalisation | No scheduled canary run                  |
| MahaRERA changes the CAPTCHA mechanism   | Medium     | Full auth path breaks                | - - token manual escape hatch                                       | —                                        |
| A registry mirror goes paywalled         | Medium     | One of three chain members lost      | _looks_paywalled; 3-way chain; roster conflicts surfaced            | —                                        |
| GST portal layout changes                | Medium     | GST sub-metric unscored              | Graded note strings; never fatal                                    | —                                        |
| Maha Bhulekh Marathi labels change       | Low        | District map misses                  | _DISTRICT_NAME_MAP with pre-rename aliases; never fuzzy-matches     | Map is manual                            |
| Anthropic API outage                     | Low        | Research + Charter degrade           | Never fatal; standalone retry commands                              | —                                        |
| Word/COM unavailable on the run host     | Medium     | <b>No PDF — the deliverable</b>      | .docx remains on disk as the fallback output                        | D5                                       |
| Template .docx corrupted or lost         | Low        | <b>Charter generation impossible</b> | Section A mandates a timestamped backup before structural change    | Backup is manual; template is gitignored |

## 26. Appendices

### Appendix A — Complete file inventory

**Application modules (29):** `main.py`·`app.py`·`config.py`·`resolver.py`·`session_auth.py`·`token_cache.py`·`api_client.py`·`discover.py`·`run_archive.py`·`promoter_portfolio.py`·`deep_research.py`·`gst_intake.py`·`gst_portal.py`·`gst_compliance.py`·`mahabhumi.py`·`cts_intake.py`·`cts_resolve.py`·`promoter_intake.py`·`attach_rera.py`·`company_charter.py`·`charter_report.py`·`charter_document.py`·`charter_research_prep.py`·`run_charter_pipeline.py`·`executive_briefing.py`·`report.py`·`build_report.py`·`finalize_report.py`

**Markdown (7):** `CLAUDE.md`·`rules.md`·`guardrails.md`·`README.md`·`CHARTER_RESHAPE_SPEC.md`·`.claude/skills/maharera-report/SKILL.md`·`output/company_charters/CHARTER_RESHAPE_CHANGE_LOG.md`

**Config / state (5):** `requirements.txt`·`.gitignore`·`source_trust_registry.json`·`.guest_token_cache.json`·`.claude/settings.local.json`

**Binary assets:** `tessdata/eng.traineddata` (4.1 MB)·`tessdata/mar.traineddata` (3.2 MB)·`RERA_Executive_Tracker.xlsx`·`docs/*.docx` (11 design artefacts)

**Tests (28):** listed in §6.2.

### Appendix B — Usage labels (the cost ledger’s key space)

`research_generate`·`verify_claim_batch`·`gap_retry_batch`·`gap_retry_verify_batch`·`charter_pass`·`material_claim_verify`·`document_grounding_verify`·`citation_completeness_judge`·`second_source_verify`·`clean_check_judge`·`citation_match`·`flag_headline`·`finding_research`·`claude_md_doc_review`·`agentic_pass` (*default*)

Each label is a separable cost line in `usage_summary.json` and in the run summary. `claude_md_doc_review` in particular is separable **by design**, so the cost of the compliance gate is visible and arguable.

As of 2026-08-13, `verify_claim_batch` / `gap_retry_batch` / `gap_retry_verify_batch` replaced the bare `verify_claim` / `gap_retry` / `gap_retry_verify` labels — one call now covers a whole chunk of sources or a whole retry round, not one source or gap. `deep_research._verify_claim` (the single-claim function) still exists and is still called directly, but only with an explicit label override (`material_claim_verify`), so the bare `verify_claim` label itself no longer appears in the ledger.

### Appendix C — Exception inventory

| Exception                         | Module                | Meaning                         |
|-----------------------------------|-----------------------|---------------------------------|
| <code>ProjectNotFoundError</code> | <code>resolver</code> | Search returned zero candidates |

| Exception               | Module                                | Meaning                                                                                                                                          |
|-------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| CaptchaTimeoutError     | session_auth,<br>gst_portal, mahabhum | Nobody solved it in time                                                                                                                         |
| BrowserClosedError      | session_auth,<br>gst_portal, mahabhum | Window closed / page unreachable                                                                                                                 |
| AmbiguousSelectionError | mahabhum                              | Carries hint and options so the caller can re-present the real list                                                                              |
| CategoryFetchError      | api_client                            | Carries .category and .status_code so callers can distinguish 401/403                                                                            |
| GstIntakeError          | gst_intake                            | Intake produced no scoreable filing data                                                                                                         |
| BudgetExhausted         | deep_research                         | A search/token budget ran dry mid-call; the “answer was malformed” and “ran out of room” cases are kept distinguishable                          |
| CostCapExceeded         | deep_research                         | Refused a call because this run’s total spend (\$16.3) is already at or over PIPELINE_COST_CAP_USD; unlike BudgetExhausted, <b>never retried</b> |
| CharterComplianceError  | company_charter                       | <b>The only class in the module.</b> Carries .results. Blocks the PDF                                                                            |
| RuntimeError            | company_charter                       | Preflight failure, External gate failure                                                                                                         |
| FileNotFoundError       | company_charter,<br>finalize_report   | Missing template / missing raw/                                                                                                                  |

#### Appendix D — Glossary

| Term               | Meaning                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Charter</b>     | The Company Charter document pair (Internal + External). The primary deliverable                                      |
| <b>Clean check</b> | A check that ran and came back clear. Under Section B it produces <b>no sentence at all</b>                           |
| <b>Gap</b>         | An open unknown the reader can act on. “ <i>We looked and could not establish this.</i> ” Never compressed or deleted |
| <b>Finding</b>     | Something established. Earns a dedicated                                                                              |

| Term                        | Meaning                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Absence</b>              | follow-up research pass<br>“We checked and it is clear.” Deleted, with three carve-outs                                  |
| <b>Flag</b>                 | A one-sentence headline in Overview & Flags ending in (Gap N). Severity: imminent / structural / monitor                 |
| <b>confirmed / observed</b> | Endpoint trust levels. confirmed = payload individually re-verified; observed = taken from real traffic, not re-verified |
| <b>Variant</b>              | internal or external — the fork at render time                                                                           |
| <b>Carryover</b>            | A pre-RERA intake result (promoter_profile / land_record) attached to a reg-no by an explicit human action               |
| <b>QRMP</b>                 | Quarterly Return Monthly Payment — the GST scheme whose due dates are state-dependent                                    |
| <b>CTS</b>                  | City Survey number — the Maharashtra land parcel identifier                                                              |
| <b>Property Card</b>        | मालमत्ता पत्रक, the Maha Bhulekh land record                                                                             |
| <b>Never fatal</b>          | A stage wrapped so that its failure logs a warning and the run continues                                                 |
| <b>Hard gate</b>            | A guard that stops the run or blocks a save/deliverable                                                                  |

#### Appendix E — Architecture decision record (retrospective)

| ADR    | Decision                                            | Rationale                                                                               | Status                 |
|--------|-----------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------|
| ADR-01 | Read the View Details href instead of clicking it   | A click fires a JS confirm() and lands on the CAPTCHA page; the href is unauthenticated | Accepted               |
| ADR-02 | Never solve a CAPTCHA programmatically              | Ethics and ToS; stated in three modules                                                 | Accepted, load-bearing |
| ADR-03 | Cache the guest token for 90 min against a ~100 min | A token must not expire mid-run                                                         | Accepted               |

| ADR    | Decision                                                                                                           | Rationale                                                                                                  | Status                                                            |
|--------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
|        | TTL                                                                                                                |                                                                                                            |                                                                   |
| ADR-04 | Per-category thread isolation with per-worker Session                                                              | requests . Session thread-safety is not guaranteed; the reuse benefit is marginal                          | Accepted                                                          |
| ADR-05 | Tier endpoints confirmed vs observed in config                                                                     | Makes the gap between “we think this works” and “it works” machine-readable and visible on the deliverable | Accepted, unusual and valuable                                    |
| ADR-06 | Put content rules in rules .md and parse them at runtime                                                           | One source of truth for both humans and prompts                                                            | Accepted                                                          |
| ADR-07 | Never send Section A to an API                                                                                     | Coding-time guidance is not content guidance; asserted by test                                             | Accepted                                                          |
| ADR-08 | Generate the External document entirely in code                                                                    | A model cannot be trusted to apply citation rules deterministically                                        | Accepted; means rule edits alone cannot change External behaviour |
| ADR-09 | Block the PDF on a semantic review, but verify every quoted violation mechanically                                 | Reduces the model’s role to “point at the text”; an invented quote cannot block                            | Accepted; the key safety argument                                 |
| ADR-10 | Keep a deterministic predecessor behind every model pass, keyed so a miss is indistinguishable from “no model ran” | A model can only match or improve, never degrade                                                           | Accepted                                                          |
| ADR-11 | Scrub the page, restore the record                                                                                 | The record must keep what the page drops                                                                   | Accepted                                                          |
| ADR-12 | Never renormalise the Developer Score                                                                              | Less disclosure <i>should</i> score lower                                                                  | Accepted                                                          |
| ADR-13 | Always renormalise Documentation Confidence                                                                        | It scores this document’s sourcing, where N/A is genuinely not                                             | Accepted                                                          |

| ADR    | Decision                                                               | Rationale                                                          | Status                                     |
|--------|------------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------|
|        |                                                                        | applicable                                                         |                                            |
| ADR-14 | Surface disagreeing director rosters rather than picking one           | Two contradictory facts beat one invented one                      | Accepted; caught a real 3-way disagreement |
| ADR-15 | Retire <code>charter_document.py</code> 's builder but keep the module | 16 symbols are still imported                                      | Accepted 2026-08-10                        |
| ADR-16 | Archive the previous run with move, not copy                           | No disk doubling; guarantees a clean target directory              | Accepted                                   |
| ADR-17 | Make GST and land-record lookups opt-in                                | Each costs a human CAPTCHA solve                                   | Accepted                                   |
| ADR-18 | Route every LLM call through one transport                             | Single place for usage accounting, JSON parsing and failure policy | Accepted                                   |

*End of Software Architecture Document.*